



October 20, 2023

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50355005

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on September 29, 2023. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

Revised Report. This report replaces the original dated 101623, it has been revised to report results down to the lab MDLs.

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

A handwritten signature in blue ink that reads "Brian Hall".

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Biosolids
Pace Project No.: 50355005

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019
Michigan Drinking Water Laboratory #9050

Ohio VAP Certified Laboratory #: CL0065
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Wisconsin Laboratory #: 999788130
USDA Foreign Soil Permit #: 525-23-13-23119
USDA Compliance Agreement #: IN-SL-22-001

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Biosolids
Pace Project No.: 50355005

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50355005001	Digester #4	Solid	09/28/23 13:55	09/29/23 10:10

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SAMPLE ANALYTE COUNT

Project: Biosolids
Pace Project No.: 50355005

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50355005001	Digester #4	EPA 8081	KAV	21	PASI-I
		EPA 8081	KAV	8	PASI-I
		EPA 8082	AM	8	PASI-I
		EPA 8151A	CPH	3	PASI-I
		EPA 6010	MTM	6	PASI-I
		EPA 6010	MTM	8	PASI-I
		EPA 6020	MGM	10	PASI-I
		EPA 7470	EAE	1	PASI-I
		EPA 7471	EAE	1	PASI-I
		EPA 8270	JCM	67	PASI-I
		EPA 8270	GRM	18	PASI-I
		EPA 8260	TMW	55	PASI-I
		EPA 5030/8260	TMW	13	PASI-I
		SM 2540G	QAK	1	PASI-I
		SM 2540G	QAK	1	PASI-I
		SM 2540G	TRK	1	PASI-I
		EPA 9038	STS	1	PASI-I
		EPA 9045	BMS	1	PASI-I
		EPA 351.2	OAS	1	PASI-I
		EPA 353.2	DAW	2	PASI-I
		SM 4500-NH3 G	OAS	1	PASI-I
		SM 4500-Cl-E	ZM	1	PASI-I
		SM 4500-CN-E	YAM	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 50355005

Sample: Digester #4 Lab ID: 50355005001 Collected: 09/28/23 13:55 Received: 09/29/23 10:10 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
8081 GCS Pesticide Solids									
Analytical Method: EPA 8081 Preparation Method: EPA 3546									
Pace Analytical Services - Indianapolis									
Aldrin	<142	ug/kg	951	142	1	10/09/23 11:08	10/12/23 19:41	309-00-2	
alpha-BHC	<138	ug/kg	951	138	1	10/09/23 11:08	10/12/23 19:41	319-84-6	
beta-BHC	<178	ug/kg	951	178	1	10/09/23 11:08	10/12/23 19:41	319-85-7	
delta-BHC	<180	ug/kg	951	180	1	10/09/23 11:08	10/12/23 19:41	319-86-8	
gamma-BHC (Lindane)	<145	ug/kg	951	145	1	10/09/23 11:08	10/12/23 19:41	58-89-9	
Chlordane (Technical)	<8820	ug/kg	19000	8820	1	10/09/23 11:08	10/12/23 19:41	57-74-9	
alpha-Chlordane	<357	ug/kg	951	357	1	10/09/23 11:08	10/12/23 19:41	5103-71-9	
gamma-Chlordane	<494	ug/kg	951	494	1	10/09/23 11:08	10/12/23 19:41	5103-74-2	
4,4'-DDD	<325	ug/kg	1900	325	1	10/09/23 11:08	10/12/23 19:41	72-54-8	
4,4'-DDE	<375	ug/kg	1900	375	1	10/09/23 11:08	10/12/23 19:41	72-55-9	
4,4'-DDT	<978	ug/kg	1900	978	1	10/09/23 11:08	10/12/23 19:41	50-29-3	
Dieldrin	<327	ug/kg	1900	327	1	10/09/23 11:08	10/12/23 19:41	60-57-1	
Endosulfan I	<168	ug/kg	951	168	1	10/09/23 11:08	10/12/23 19:41	959-98-8	
Endosulfan II	<704	ug/kg	1900	704	1	10/09/23 11:08	10/12/23 19:41	33213-65-9	
Endosulfan sulfate	<359	ug/kg	1900	359	1	10/09/23 11:08	10/12/23 19:41	1031-07-8	
Endrin	<300	ug/kg	1900	300	1	10/09/23 11:08	10/12/23 19:41	72-20-8	
Endrin aldehyde	<567	ug/kg	1900	567	1	10/09/23 11:08	10/12/23 19:41	7421-93-4	
Heptachlor	<181	ug/kg	951	181	1	10/09/23 11:08	10/12/23 19:41	76-44-8	
Heptachlor epoxide	<157	ug/kg	951	157	1	10/09/23 11:08	10/12/23 19:41	1024-57-3	
Toxaphene	<5520	ug/kg	19000	5520	1	10/09/23 11:08	10/12/23 19:41	8001-35-2	
Surrogates									
Decachlorobiphenyl (S)	65	%.	10-166		1	10/09/23 11:08	10/12/23 19:41	2051-24-3	
8081 GCS Pest RV, TCLP									
Analytical Method: EPA 8081 Preparation Method: EPA 3510									
Leachate Method/Date: EPA 1311; 10/04/23 16:30 Initial pH: 8.67; Final pH: 5.13									
Pace Analytical Services - Indianapolis									
gamma-BHC (Lindane)	<0.00025	mg/L	0.00025	0.00025	1	10/09/23 17:23	10/10/23 15:49	58-89-9	
Chlordane (Technical)	<0.0050	mg/L	0.0050	0.0050	1	10/09/23 17:23	10/10/23 15:49	57-74-9	
Endrin	<0.00050	mg/L	0.00050	0.00050	1	10/09/23 17:23	10/10/23 15:49	72-20-8	
Heptachlor	<0.00025	mg/L	0.00025	0.00025	1	10/09/23 17:23	10/10/23 15:49	76-44-8	
Heptachlor epoxide	<0.00025	mg/L	0.00025	0.00025	1	10/09/23 17:23	10/10/23 15:49	1024-57-3	
Methoxychlor	<0.0025	mg/L	0.0025	0.0025	1	10/09/23 17:23	10/10/23 15:49	72-43-5	
Toxaphene	<0.0050	mg/L	0.0050	0.0050	1	10/09/23 17:23	10/10/23 15:49	8001-35-2	
Surrogates									
Decachlorobiphenyl (S)	52	%.	10-124		1	10/09/23 17:23	10/10/23 15:49	2051-24-3	
8082 PCB Solids									
Analytical Method: EPA 8082 Preparation Method: EPA 3546									
Pace Analytical Services - Indianapolis									
PCB-1016 (Aroclor 1016)	<618	ug/kg	16700	618	1	10/01/23 14:09	10/03/23 00:19	12674-11-2	
PCB-1221 (Aroclor 1221)	<782	ug/kg	16700	782	1	10/01/23 14:09	10/03/23 00:19	11104-28-2	
PCB-1232 (Aroclor 1232)	<867	ug/kg	16700	867	1	10/01/23 14:09	10/03/23 00:19	11141-16-5	
PCB-1242 (Aroclor 1242)	<730	ug/kg	16700	730	1	10/01/23 14:09	10/03/23 00:19	53469-21-9	
PCB-1248 (Aroclor 1248)	<512	ug/kg	16700	512	1	10/01/23 14:09	10/03/23 00:19	12672-29-6	
PCB-1254 (Aroclor 1254)	<750	ug/kg	16700	750	1	10/01/23 14:09	10/03/23 00:19	11097-69-1	

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 50355005

Sample: Digester #4 Lab ID: 50355005001 Collected: 09/28/23 13:55 Received: 09/29/23 10:10 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
8082 PCB Solids									
Analytical Method: EPA 8082 Preparation Method: EPA 3546									
Pace Analytical Services - Indianapolis									
PCB-1260 (Aroclor 1260)	<783	ug/kg	16700	783	1	10/01/23 14:09	10/03/23 00:19	11096-82-5	
Surrogates									
Tetrachloro-m-xylene (S)	43	%.	10-133		1	10/01/23 14:09	10/03/23 00:19	877-09-8	
8151A Cl Acid Herbicides TCLP									
Analytical Method: EPA 8151A Preparation Method: EPA 8151									
Leachate Method/Date: EPA 1311; 10/04/23 16:30 Initial pH: 8.67; Final pH: 5.13									
Pace Analytical Services - Indianapolis									
2,4-D	<0.0050	mg/L	0.0050	0.0050	1	10/09/23 14:05	10/10/23 23:33	94-75-7	
2,4,5-TP (Silvex)	<0.0050	mg/L	0.0050	0.0050	1	10/09/23 14:05	10/10/23 23:33	93-72-1	
Surrogates									
2,4-DCAA (S)	51	%.	22-132		1	10/09/23 14:05	10/10/23 23:33	19719-28-9	
6010 MET ICP									
Analytical Method: EPA 6010 Preparation Method: EPA 3050									
Pace Analytical Services - Indianapolis									
Aluminum	8010	mg/kg	4700	101	1	10/03/23 15:40	10/12/23 14:10	7429-90-5	
Calcium	37400	mg/kg	4700	411	1	10/03/23 15:40	10/12/23 14:10	7440-70-2	
Magnesium	8270	mg/kg	4700	50.4	1	10/03/23 15:40	10/12/23 14:10	7439-95-4	
Phosphorus	41400	mg/kg	4700	510	1	10/03/23 15:40	10/12/23 14:10	7723-14-0	N2
Potassium	7740	mg/kg	4700	788	1	10/03/23 15:40	10/12/23 14:10	7440-09-7	
Sodium	11000	mg/kg	4700	244	1	10/03/23 15:40	10/12/23 14:10	7440-23-5	
6010 MET ICP, TCLP									
Analytical Method: EPA 6010 Preparation Method: EPA 3010									
Leachate Method/Date: EPA 1311; 10/04/23 16:30 Initial pH: 8.67; Final pH: 5.13									
Pace Analytical Services - Indianapolis									
Arsenic	<0.050	mg/L	0.10	0.050	1	10/05/23 15:42	10/06/23 13:05	7440-38-2	
Barium	<0.25	mg/L	5.0	0.25	1	10/05/23 15:42	10/06/23 13:05	7440-39-3	
Cadmium	<0.025	mg/L	0.050	0.025	1	10/05/23 15:42	10/06/23 13:05	7440-43-9	
Chromium	0.14	mg/L	0.10	0.052	1	10/05/23 15:42	10/06/23 13:05	7440-47-3	
Lead	<0.050	mg/L	0.10	0.050	1	10/05/23 15:42	10/06/23 13:05	7439-92-1	
Selenium	<0.050	mg/L	0.10	0.050	1	10/05/23 15:42	10/06/23 13:05	7782-49-2	
Silver	<0.050	mg/L	0.10	0.050	1	10/05/23 15:42	10/06/23 13:05	7440-22-4	
Zinc	1.1	mg/L	0.20	0.10	1	10/05/23 15:42	10/06/23 13:05	7440-66-6	
6020 MET ICPMS									
Analytical Method: EPA 6020 Preparation Method: EPA 3050B									
Pace Analytical Services - Indianapolis									
Arsenic	30.4	mg/kg	9.2	0.87	1	10/06/23 09:00	10/11/23 17:12	7440-38-2	
Cadmium	5.2	mg/kg	4.6	2.1	1	10/06/23 09:00	10/11/23 17:12	7440-43-9	
Chromium	166	mg/kg	18.4	2.3	1	10/06/23 09:00	10/11/23 17:12	7440-47-3	
Copper	1060	mg/kg	9.2	3.7	1	10/06/23 09:00	10/11/23 17:12	7440-50-8	
Lead	24.7	mg/kg	9.2	0.36	1	10/06/23 09:00	10/11/23 17:12	7439-92-1	
Molybdenum	29.2	mg/kg	9.2	0.85	1	10/06/23 09:00	10/11/23 17:12	7439-98-7	N2
Nickel	36.1	mg/kg	4.6	1.2	1	10/06/23 09:00	10/11/23 17:12	7440-02-0	
Selenium	7.4J	mg/kg	9.2	2.6	1	10/06/23 09:00	10/11/23 17:12	7782-49-2	
Silver	3.9J	mg/kg	4.6	0.20	1	10/06/23 09:00	10/11/23 17:12	7440-22-4	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 50355005

Sample: Digester #4 Lab ID: 50355005001 Collected: 09/28/23 13:55 Received: 09/29/23 10:10 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
6020 MET ICPMS									
Analytical Method: EPA 6020 Preparation Method: EPA 3050B									
Pace Analytical Services - Indianapolis									
Zinc	1430	mg/kg	46.0	5.8	1	10/06/23 09:00	10/11/23 17:12	7440-66-6	
7470 Mercury, TCLP									
Analytical Method: EPA 7470 Preparation Method: EPA 7470									
Leachate Method/Date: EPA 1311; 10/04/23 16:30 Initial pH: 8.67; Final pH: 5.13									
Pace Analytical Services - Indianapolis									
Mercury	<0.0010	mg/L	0.0020	0.0010	1	10/05/23 13:08	10/05/23 20:03	7439-97-6	
7471 Mercury									
Analytical Method: EPA 7471 Preparation Method: EPA 7471									
Pace Analytical Services - Indianapolis									
Mercury	<1.8	mg/kg	18.7	1.8	1	10/10/23 12:23	10/10/23 16:38	7439-97-6	
8270 SVOC SS Soil									
Analytical Method: EPA 8270 Preparation Method: EPA 3546									
Pace Analytical Services - Indianapolis									
Acenaphthene	<87200	ug/kg	326000	87200	1	10/03/23 11:50	10/04/23 23:20	83-32-9	
Acenaphthylene	<97600	ug/kg	326000	97600	1	10/03/23 11:50	10/04/23 23:20	208-96-8	
Anthracene	<134000	ug/kg	326000	134000	1	10/03/23 11:50	10/04/23 23:20	120-12-7	
Benzo(a)anthracene	<96900	ug/kg	326000	96900	1	10/03/23 11:50	10/04/23 23:20	56-55-3	L1
Benzo(a)pyrene	<107000	ug/kg	326000	107000	1	10/03/23 11:50	10/04/23 23:20	50-32-8	L1
Benzo(b)fluoranthene	<107000	ug/kg	326000	107000	1	10/03/23 11:50	10/04/23 23:20	205-99-2	L1
Benzo(g,h,i)perylene	<119000	ug/kg	326000	119000	1	10/03/23 11:50	10/04/23 23:20	191-24-2	
Benzo(k)fluoranthene	<118000	ug/kg	326000	118000	1	10/03/23 11:50	10/04/23 23:20	207-08-9	
Benzyl alcohol	<105000	ug/kg	653000	105000	1	10/03/23 11:50	10/04/23 23:20	100-51-6	
4-Bromophenylphenyl ether	<122000	ug/kg	326000	122000	1	10/03/23 11:50	10/04/23 23:20	101-55-3	
Butylbenzylphthalate	<179000	ug/kg	326000	179000	1	10/03/23 11:50	10/04/23 23:20	85-68-7	
4-Chloro-3-methylphenol	<133000	ug/kg	653000	133000	1	10/03/23 11:50	10/04/23 23:20	59-50-7	
4-Chloroaniline	<85900	ug/kg	653000	85900	1	10/03/23 11:50	10/04/23 23:20	106-47-8	
bis(2-Chloroethoxy)methane	<104000	ug/kg	326000	104000	1	10/03/23 11:50	10/04/23 23:20	111-91-1	
bis(2-Chloroethyl) ether	<126000	ug/kg	326000	126000	1	10/03/23 11:50	10/04/23 23:20	111-44-4	
bis(2chloro1methylethyl) ether	<114000	ug/kg	326000	114000	1	10/03/23 11:50	10/04/23 23:20	108-60-1	
2-Chloronaphthalene	<91700	ug/kg	326000	91700	1	10/03/23 11:50	10/04/23 23:20	91-58-7	
2-Chlorophenol	<117000	ug/kg	326000	117000	1	10/03/23 11:50	10/04/23 23:20	95-57-8	
4-Chlorophenylphenyl ether	<101000	ug/kg	326000	101000	1	10/03/23 11:50	10/04/23 23:20	7005-72-3	
Chrysene	<107000	ug/kg	326000	107000	1	10/03/23 11:50	10/04/23 23:20	218-01-9	
Dibenz(a,h)anthracene	<117000	ug/kg	326000	117000	1	10/03/23 11:50	10/04/23 23:20	53-70-3	
Dibenzofuran	<102000	ug/kg	326000	102000	1	10/03/23 11:50	10/04/23 23:20	132-64-9	
3,3'-Dichlorobenzidine	<324000	ug/kg	653000	324000	1	10/03/23 11:50	10/04/23 23:20	91-94-1	
2,4-Dichlorophenol	<113000	ug/kg	326000	113000	1	10/03/23 11:50	10/04/23 23:20	120-83-2	
Diethylphthalate	<275000	ug/kg	326000	275000	1	10/03/23 11:50	10/04/23 23:20	84-66-2	
2,4-Dimethylphenol	<113000	ug/kg	326000	113000	1	10/03/23 11:50	10/04/23 23:20	105-67-9	
Dimethylphthalate	<109000	ug/kg	326000	109000	1	10/03/23 11:50	10/04/23 23:20	131-11-3	
Di-n-butylphthalate	<119000	ug/kg	326000	119000	1	10/03/23 11:50	10/04/23 23:20	84-74-2	
4,6-Dinitro-2-methylphenol	<199000	ug/kg	653000	199000	1	10/03/23 11:50	10/04/23 23:20	534-52-1	
2,4-Dinitrophenol	<180000	ug/kg	1580000	180000	1	10/03/23 11:50	10/04/23 23:20	51-28-5	
2,4-Dinitrotoluene	<109000	ug/kg	326000	109000	1	10/03/23 11:50	10/04/23 23:20	121-14-2	

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 50355005

Sample: Digester #4 Lab ID: 50355005001 Collected: 09/28/23 13:55 Received: 09/29/23 10:10 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil									
Analytical Method: EPA 8270 Preparation Method: EPA 3546									
Pace Analytical Services - Indianapolis									
2,6-Dinitrotoluene	<93300	ug/kg	326000	93300	1	10/03/23 11:50	10/04/23 23:20	606-20-2	
Di-n-octylphthalate	<119000	ug/kg	326000	119000	1	10/03/23 11:50	10/04/23 23:20	117-84-0	
bis(2-Ethylhexyl)phthalate	<99900	ug/kg	326000	99900	1	10/03/23 11:50	10/04/23 23:20	117-81-7	
Fluoranthene	<125000	ug/kg	326000	125000	1	10/03/23 11:50	10/04/23 23:20	206-44-0	
Fluorene	<110000	ug/kg	326000	110000	1	10/03/23 11:50	10/04/23 23:20	86-73-7	
Hexachloro-1,3-butadiene	<88600	ug/kg	326000	88600	1	10/03/23 11:50	10/04/23 23:20	87-68-3	
Hexachlorobenzene	<82600	ug/kg	326000	82600	1	10/03/23 11:50	10/04/23 23:20	118-74-1	
Hexachlorocyclopentadiene	<160000	ug/kg	326000	160000	1	10/03/23 11:50	10/04/23 23:20	77-47-4	
Hexachloroethane	<99200	ug/kg	326000	99200	1	10/03/23 11:50	10/04/23 23:20	67-72-1	
Indeno(1,2,3-cd)pyrene	<118000	ug/kg	326000	118000	1	10/03/23 11:50	10/04/23 23:20	193-39-5	
Isophorone	<106000	ug/kg	326000	106000	1	10/03/23 11:50	10/04/23 23:20	78-59-1	
1-Methylnaphthalene	<119000	ug/kg	326000	119000	1	10/03/23 11:50	10/04/23 23:20	90-12-0	
2-Methylnaphthalene	<97400	ug/kg	326000	97400	1	10/03/23 11:50	10/04/23 23:20	91-57-6	
2-Methylphenol(o-Cresol)	<139000	ug/kg	326000	139000	1	10/03/23 11:50	10/04/23 23:20	95-48-7	
3&4-Methylphenol(m&p Cresol)	<136000	ug/kg	653000	136000	1	10/03/23 11:50	10/04/23 23:20		
Naphthalene	<93800	ug/kg	326000	93800	1	10/03/23 11:50	10/04/23 23:20	91-20-3	
2-Nitroaniline	<134000	ug/kg	326000	134000	1	10/03/23 11:50	10/04/23 23:20	88-74-4	
3-Nitroaniline	<118000	ug/kg	326000	118000	1	10/03/23 11:50	10/04/23 23:20	99-09-2	
4-Nitroaniline	<130000	ug/kg	326000	130000	1	10/03/23 11:50	10/04/23 23:20	100-01-6	
Nitrobenzene	<109000	ug/kg	326000	109000	1	10/03/23 11:50	10/04/23 23:20	98-95-3	
2-Nitrophenol	<126000	ug/kg	326000	126000	1	10/03/23 11:50	10/04/23 23:20	88-75-5	
4-Nitrophenol	<247000	ug/kg	1580000	247000	1	10/03/23 11:50	10/04/23 23:20	100-02-7	
N-Nitroso-di-n-propylamine	<125000	ug/kg	326000	125000	1	10/03/23 11:50	10/04/23 23:20	621-64-7	
N-Nitrosodiphenylamine	<106000	ug/kg	326000	106000	1	10/03/23 11:50	10/04/23 23:20	86-30-6	
Pentachlorophenol	<251000	ug/kg	1580000	251000	1	10/03/23 11:50	10/04/23 23:20	87-86-5	
Phenanthrene	<127000	ug/kg	326000	127000	1	10/03/23 11:50	10/04/23 23:20	85-01-8	
Phenol	<119000	ug/kg	326000	119000	1	10/03/23 11:50	10/04/23 23:20	108-95-2	
Pyrene	<103000	ug/kg	326000	103000	1	10/03/23 11:50	10/04/23 23:20	129-00-0	
2,4,5-Trichlorophenol	<114000	ug/kg	326000	114000	1	10/03/23 11:50	10/04/23 23:20	95-95-4	
2,4,6-Trichlorophenol	<100000	ug/kg	326000	100000	1	10/03/23 11:50	10/04/23 23:20	88-06-2	
Surrogates									
Nitrobenzene-d5 (S)	72	%	32-95		1	10/03/23 11:50	10/04/23 23:20	4165-60-0	
Phenol-d5 (S)	82	%	27-116		1	10/03/23 11:50	10/04/23 23:20	4165-62-2	
2-Fluorophenol (S)	79	%	21-109		1	10/03/23 11:50	10/04/23 23:20	367-12-4	
2,4,6-Tribromophenol (S)	91	%	10-121		1	10/03/23 11:50	10/04/23 23:20	118-79-6	
2-Fluorobiphenyl (S)	80	%	33-102		1	10/03/23 11:50	10/04/23 23:20	321-60-8	
p-Terphenyl-d14 (S)	93	%	20-120		1	10/03/23 11:50	10/04/23 23:20	1718-51-0	

8270 MSSV TCLP Sep Funnel

Analytical Method: EPA 8270 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 10/04/23 16:30 Initial pH: 8.67; Final pH: 5.13

Pace Analytical Services - Indianapolis

1,4-Dichlorobenzene	<0.050	mg/L	0.10	0.050	1	10/10/23 10:11	10/10/23 19:03	106-46-7	
2,4-Dinitrotoluene	<0.050	mg/L	0.10	0.050	1	10/10/23 10:11	10/10/23 19:03	121-14-2	
Hexachloro-1,3-butadiene	<0.050	mg/L	0.10	0.050	1	10/10/23 10:11	10/10/23 19:03	87-68-3	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 50355005

Sample: Digester #4 Lab ID: 50355005001 Collected: 09/28/23 13:55 Received: 09/29/23 10:10 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
8270 MSSV TCLP Sep Funnel									
Analytical Method: EPA 8270 Preparation Method: EPA 3510									
Leachate Method/Date: EPA 1311; 10/04/23 16:30 Initial pH: 8.67; Final pH: 5.13									
Pace Analytical Services - Indianapolis									
Hexachlorobenzene	<0.050	mg/L	0.10	0.050	1	10/10/23 10:11	10/10/23 19:03	118-74-1	
Hexachloroethane	<0.050	mg/L	0.10	0.050	1	10/10/23 10:11	10/10/23 19:03	67-72-1	
2-Methylphenol(o-Cresol)	<0.050	mg/L	0.10	0.050	1	10/10/23 10:11	10/10/23 19:03	95-48-7	
3&4-Methylphenol(m&p Cresol)	<0.10	mg/L	0.20	0.10	1	10/10/23 10:11	10/10/23 19:03		
Nitrobenzene	<0.050	mg/L	0.10	0.050	1	10/10/23 10:11	10/10/23 19:03	98-95-3	M1
Pentachlorophenol	<0.25	mg/L	0.50	0.25	1	10/10/23 10:11	10/10/23 19:03	87-86-5	
Pyridine	<0.10	mg/L	0.10	0.10	1	10/10/23 10:11	10/10/23 19:03	110-86-1	
2,4,5-Trichlorophenol	<0.050	mg/L	0.50	0.050	1	10/10/23 10:11	10/10/23 19:03	95-95-4	
2,4,6-Trichlorophenol	<0.050	mg/L	0.10	0.050	1	10/10/23 10:11	10/10/23 19:03	88-06-2	
Surrogates									
Nitrobenzene-d5 (S)	0	%	44-107		1	10/10/23 10:11	10/10/23 19:03	4165-60-0	S0
2-Fluorobiphenyl (S)	65	%	36-92		1	10/10/23 10:11	10/10/23 19:03	321-60-8	
p-Terphenyl-d14 (S)	88	%	57-146		1	10/10/23 10:11	10/10/23 19:03	1718-51-0	
Phenol-d5 (S)	17	%	15-47		1	10/10/23 10:11	10/10/23 19:03	4165-62-2	
2-Fluorophenol (S)	51	%	23-69		1	10/10/23 10:11	10/10/23 19:03	367-12-4	
2,4,6-Tribromophenol (S)	114	%	37-127		1	10/10/23 10:11	10/10/23 19:03	118-79-6	
8260 MSV 5030 Low Level									
Analytical Method: EPA 8260									
Pace Analytical Services - Indianapolis									
Acetone	6.9J	mg/kg	9.9	0.59	1		10/04/23 15:05	67-64-1	
Acrolein	<1.7	mg/kg	9.9	1.7	1		10/04/23 15:05	107-02-8	L1
Acrylonitrile	<0.20	mg/kg	9.9	0.20	1		10/04/23 15:05	107-13-1	
Benzene	<0.048	mg/kg	0.49	0.048	1		10/04/23 15:05	71-43-2	
Bromodichloromethane	<0.031	mg/kg	0.49	0.031	1		10/04/23 15:05	75-27-4	
Bromoform	<0.030	mg/kg	0.49	0.030	1		10/04/23 15:05	75-25-2	
Bromomethane	<0.17	mg/kg	0.49	0.17	1		10/04/23 15:05	74-83-9	
2-Butanone (MEK)	1.7J	mg/kg	2.5	0.33	1		10/04/23 15:05	78-93-3	
Carbon disulfide	0.36J	mg/kg	0.99	0.038	1		10/04/23 15:05	75-15-0	
Carbon tetrachloride	<0.16	mg/kg	0.49	0.16	1		10/04/23 15:05	56-23-5	
Chlorobenzene	<0.036	mg/kg	0.49	0.036	1		10/04/23 15:05	108-90-7	
Chloroethane	<0.098	mg/kg	0.49	0.098	1		10/04/23 15:05	75-00-3	L1
Chloroform	<0.32	mg/kg	0.49	0.32	1		10/04/23 15:05	67-66-3	
Chloromethane	<0.046	mg/kg	0.49	0.046	1		10/04/23 15:05	74-87-3	L1
1,2-Dibromo-3-chloropropane	<0.11	mg/kg	0.99	0.11	1		10/04/23 15:05	96-12-8	
Dibromochloromethane	<0.028	mg/kg	0.49	0.028	1		10/04/23 15:05	124-48-1	
1,2-Dibromoethane (EDB)	<0.035	mg/kg	0.49	0.035	1		10/04/23 15:05	106-93-4	
Dibromomethane	<0.036	mg/kg	0.49	0.036	1		10/04/23 15:05	74-95-3	
1,2-Dichlorobenzene	<0.039	mg/kg	0.49	0.039	1		10/04/23 15:05	95-50-1	
1,3-Dichlorobenzene	<0.043	mg/kg	0.49	0.043	1		10/04/23 15:05	541-73-1	
1,4-Dichlorobenzene	<0.043	mg/kg	0.49	0.043	1		10/04/23 15:05	106-46-7	
trans-1,4-Dichloro-2-butene	<0.055	mg/kg	9.9	0.055	1		10/04/23 15:05	110-57-6	
Dichlorodifluoromethane	<0.033	mg/kg	0.49	0.033	1		10/04/23 15:05	75-71-8	
1,1-Dichloroethane	<0.034	mg/kg	0.49	0.034	1		10/04/23 15:05	75-34-3	

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 50355005

Sample: Digester #4 Lab ID: 50355005001 Collected: 09/28/23 13:55 Received: 09/29/23 10:10 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level									
Analytical Method: EPA 8260									
Pace Analytical Services - Indianapolis									
1,2-Dichloroethane	<0.032	mg/kg	0.49	0.032	1		10/04/23 15:05	107-06-2	
1,1-Dichloroethene	<0.035	mg/kg	0.49	0.035	1		10/04/23 15:05	75-35-4	
cis-1,2-Dichloroethene	<0.036	mg/kg	0.49	0.036	1		10/04/23 15:05	156-59-2	
trans-1,2-Dichloroethene	<0.038	mg/kg	0.49	0.038	1		10/04/23 15:05	156-60-5	
1,2-Dichloropropane	<0.036	mg/kg	0.49	0.036	1		10/04/23 15:05	78-87-5	
cis-1,3-Dichloropropene	<0.034	mg/kg	0.49	0.034	1		10/04/23 15:05	10061-01-5	
trans-1,3-Dichloropropene	<0.034	mg/kg	0.49	0.034	1		10/04/23 15:05	10061-02-6	
Ethylbenzene	<0.099	mg/kg	0.49	0.099	1		10/04/23 15:05	100-41-4	
Ethyl methacrylate	<0.037	mg/kg	9.9	0.037	1		10/04/23 15:05	97-63-2	
2-Hexanone	<0.22	mg/kg	9.9	0.22	1		10/04/23 15:05	591-78-6	
Iodomethane	<0.16	mg/kg	9.9	0.16	1		10/04/23 15:05	74-88-4	
Methylene Chloride	<1.4	mg/kg	2.0	1.4	1		10/04/23 15:05	75-09-2	
Methyl methacrylate	<0.050	mg/kg	9.9	0.050	1		10/04/23 15:05	80-62-6	
4-Methyl-2-pentanone (MIBK)	<0.21	mg/kg	2.5	0.21	1		10/04/23 15:05	108-10-1	
Methyl-tert-butyl ether	<0.034	mg/kg	0.49	0.034	1		10/04/23 15:05	1634-04-4	
Styrene	<0.050	mg/kg	0.49	0.050	1		10/04/23 15:05	100-42-5	
1,1,1,2-Tetrachloroethane	<0.034	mg/kg	0.49	0.034	1		10/04/23 15:05	630-20-6	
1,1,2,2-Tetrachloroethane	<0.037	mg/kg	0.49	0.037	1		10/04/23 15:05	79-34-5	
Tetrachloroethene	<0.037	mg/kg	0.49	0.037	1		10/04/23 15:05	127-18-4	
Toluene	0.077J	mg/kg	0.49	0.045	1		10/04/23 15:05	108-88-3	
1,1,1-Trichloroethane	<0.034	mg/kg	0.49	0.034	1		10/04/23 15:05	71-55-6	
1,1,2-Trichloroethane	<0.037	mg/kg	0.49	0.037	1		10/04/23 15:05	79-00-5	
Trichloroethene	<0.043	mg/kg	0.49	0.043	1		10/04/23 15:05	79-01-6	
Trichlorofluoromethane	<0.034	mg/kg	0.49	0.034	1		10/04/23 15:05	75-69-4	
1,2,3-Trichloropropane	<0.038	mg/kg	0.49	0.038	1		10/04/23 15:05	96-18-4	
Vinyl acetate	<0.18	mg/kg	9.9	0.18	1		10/04/23 15:05	108-05-4	
Vinyl chloride	<0.036	mg/kg	0.49	0.036	1		10/04/23 15:05	75-01-4	
Xylene (Total)	<0.38	mg/kg	0.99	0.38	1		10/04/23 15:05	1330-20-7	
Surrogates									
Dibromofluoromethane (S)	103	%	75-135		1		10/04/23 15:05	1868-53-7	
Toluene-d8 (S)	105	%	65-148		1		10/04/23 15:05	2037-26-5	
4-Bromofluorobenzene (S)	89	%	63-132		1		10/04/23 15:05	460-00-4	

8260 MSV TCLP

Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 10/10/23 22:30

Pace Analytical Services - Indianapolis

Benzene	<0.010	mg/L	0.050	0.010	1		10/12/23 20:46	71-43-2	
2-Butanone (MEK)	<0.50	mg/L	1.0	0.50	1		10/12/23 20:46	78-93-3	
Carbon tetrachloride	<0.025	mg/L	0.050	0.025	1		10/12/23 20:46	56-23-5	
Chlorobenzene	<0.025	mg/L	0.050	0.025	1		10/12/23 20:46	108-90-7	
Chloroform	<0.025	mg/L	0.050	0.025	1		10/12/23 20:46	67-66-3	
1,2-Dichloroethane	<0.025	mg/L	0.050	0.025	1		10/12/23 20:46	107-06-2	
1,1-Dichloroethene	<0.025	mg/L	0.050	0.025	1		10/12/23 20:46	75-35-4	
Tetrachloroethene	<0.025	mg/L	0.050	0.025	1		10/12/23 20:46	127-18-4	
Trichloroethene	<0.025	mg/L	0.050	0.025	1		10/12/23 20:46	79-01-6	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50355005

Sample: Digester #4 Lab ID: 50355005001 Collected: 09/28/23 13:55 Received: 09/29/23 10:10 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV TCLP Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 10/10/23 22:30 Pace Analytical Services - Indianapolis									
Vinyl chloride	<0.010	mg/L	0.020	0.010	1		10/12/23 20:46	75-01-4	
Surrogates									
4-Bromofluorobenzene (S)	101	%	79-124		1		10/12/23 20:46	460-00-4	
Dibromofluoromethane (S)	105	%	82-128		1		10/12/23 20:46	1868-53-7	
Toluene-d8 (S)	100	%	73-122		1		10/12/23 20:46	2037-26-5	
Percent Moisture Analytical Method: SM 2540G Pace Analytical Services - Indianapolis									
Percent Moisture	99.0	%	0.10	0.10	1		10/04/23 18:23		N2
2540G Total Percent Solids Analytical Method: SM 2540G Pace Analytical Services - Indianapolis									
Total Solids	1.0	%	0.10	0.10	1		10/04/23 18:23		N2
2540G Total Volatile Solids Analytical Method: SM 2540G Pace Analytical Services - Indianapolis									
Total Volatile Solids	54.1	%	0.10	0.10	1		10/04/23 18:23		N2
9038 Sulfate Soil Analytical Method: EPA 9038 Preparation Method: EPA 9038 Pace Analytical Services - Indianapolis									
Sulfate	7640J	mg/kg	9890	4940	1	10/05/23 13:15	10/05/23 15:23	14808-79-8	N2
9045 pH Soil Analytical Method: EPA 9045 Pace Analytical Services - Indianapolis									
pH at 25 Degrees C	<0.10	Std. Units	0.10	0.10	1		10/12/23 09:58		H3,PO
351.2 Total Kjeldahl Nitrogen Analytical Method: EPA 351.2 Preparation Method: EPA 351.2 Pace Analytical Services - Indianapolis									
Nitrogen, Kjeldahl, Total	130000	mg/kg	9700	6860	2	10/11/23 07:46	10/11/23 13:01	7727-37-9	N2
353.2 Nitrogen, NO2/NO3 Analytical Method: EPA 353.2 Preparation Method: EPA 353.2 Pace Analytical Services - Indianapolis									
Nitrogen, NO2 plus NO3	<245	mg/kg	490	245	1	10/09/23 23:03	10/10/23 20:52		N2
Nitrogen, Nitrate	<245	mg/kg	490	245	1	10/09/23 23:03	10/10/23 20:52	14797-55-8	N2
4500 Ammonia Soil, Distilled Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B Pace Analytical Services - Indianapolis									
Nitrogen, Ammonia	98400	mg/kg	4340	1990	1	10/11/23 11:50	10/11/23 17:56	7664-41-7	N2
4500 Chloride in Soil Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E Pace Analytical Services - Indianapolis									
Chloride	15200	mg/kg	9860	4930	1	10/12/23 19:30	10/13/23 13:52	16887-00-6	N2

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50355005

Sample: Digester #4 Lab ID: 50355005001 Collected: 09/28/23 13:55 Received: 09/29/23 10:10 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
4500CNE Cyanide, Soil									
Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis									
Cyanide	<25.1	mg/kg	47.5	25.1	1	10/12/23 09:40	10/12/23 12:11	57-12-5	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50355005

QC Batch:	755883	Analysis Method:	EPA 7470
QC Batch Method:	EPA 7470	Analysis Description:	7470 Mercury TCLP
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3464237 Matrix: Water
Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	mg/L	<0.00033	0.00067	0.00033	10/05/23 19:19	

LABORATORY CONTROL SAMPLE: 3464238

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.005	0.0054	107	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3464239 3464240

Parameter	Units	92690877002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/L	ND	0.015	0.015	0.015	0.015	101	100	75-125	1	20	

MATRIX SPIKE SAMPLE: 3464241

Parameter	Units	92690213001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.016	106	75-125	

MATRIX SPIKE SAMPLE: 3464242

Parameter	Units	92690214001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	103	75-125	

MATRIX SPIKE SAMPLE: 3464243

Parameter	Units	50355058001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.15	0.078	52	75-125	M0

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50355005

MATRIX SPIKE SAMPLE:		3464244					
Parameter	Units	50355253001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	101	75-125	

MATRIX SPIKE SAMPLE:		3464245					
Parameter	Units	50355005001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	<0.0010	0.015	0.015	101	75-125	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 756431

Analysis Method: EPA 7471

QC Batch Method: EPA 7471

Analysis Description: 7471 Mercury

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3466780

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	mg/kg	<0.019	0.20	0.019	10/10/23 16:33	

LABORATORY CONTROL SAMPLE: 3466781

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.5	0.48	95	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3466782 3466783

Parameter	Units	50355076010 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.65	0.65	0.64	0.63	98	97	75-125	1	20	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 755107

Analysis Method: EPA 6010

QC Batch Method: EPA 3050

Analysis Description: 6010 MET

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3460621

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Aluminum	mg/kg	3.3J	50.0	1.1	10/12/23 11:29	
Calcium	mg/kg	<4.4	50.0	4.4	10/12/23 11:29	
Magnesium	mg/kg	0.54J	50.0	0.54	10/12/23 11:29	
Phosphorus	mg/kg	<5.4	50.0	5.4	10/12/23 11:29	N2
Potassium	mg/kg	<8.4	50.0	8.4	10/12/23 11:29	
Sodium	mg/kg	<2.6	50.0	2.6	10/12/23 11:29	

LABORATORY CONTROL SAMPLE: 3460622

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	500	503	101	80-120	
Calcium	mg/kg	500	505	101	80-120	
Magnesium	mg/kg	500	498	100	80-120	
Phosphorus	mg/kg	50	50.8	102	80-120	N2
Potassium	mg/kg	500	501	100	80-120	
Sodium	mg/kg	500	499	100	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3460623 3460624

Parameter	Units	50355022004 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	5710	506	530	8390	8660	531	558	75-125	3	20	P6
Calcium	mg/kg	1450	506	530	2500	2210	206	143	75-125	12	20	M3
Magnesium	mg/kg	997	506	530	1690	1790	136	150	75-125	6	20	M3
Phosphorus	mg/kg	216	50.6	53	271	265	109	93	75-125	2	20	N2
Potassium	mg/kg	451	506	530	1090	1180	127	138	75-125	8	20	2d,M3
Sodium	mg/kg	ND	506	530	483	540	89	96	75-125	11	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 755887

Analysis Method: EPA 6010

QC Batch Method: EPA 3010

Analysis Description: 6010 MET TCLP

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3464277

Matrix: Water

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Arsenic	mg/L	<0.0050	0.010	0.0050	10/06/23 12:42	
Barium	mg/L	<0.025	0.50	0.025	10/06/23 12:42	
Cadmium	mg/L	<0.0025	0.0050	0.0025	10/06/23 12:42	
Chromium	mg/L	<0.0052	0.010	0.0052	10/06/23 12:42	
Lead	mg/L	<0.0050	0.010	0.0050	10/06/23 12:42	
Selenium	mg/L	<0.0050	0.010	0.0050	10/06/23 12:42	
Silver	mg/L	<0.0050	0.010	0.0050	10/06/23 12:42	
Zinc	mg/L	<0.010	0.020	0.010	10/06/23 12:42	

LABORATORY CONTROL SAMPLE: 3464278

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	1	1.0	103	80-120	
Barium	mg/L	1	1.0	100	80-120	
Cadmium	mg/L	1	0.97	97	80-120 F5	
Chromium	mg/L	1	1.0	101	80-120	
Lead	mg/L	1	0.93	93	80-120	
Selenium	mg/L	1	1.0	100	80-120	
Silver	mg/L	0.5	0.51	101	80-120	
Zinc	mg/L	1	0.98	98	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3464279

3464280

Parameter	Units	50354152001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/L	ND	10	10	11.1	10.9	111	109	50-150	2	20	
Barium	mg/L	ND	10	10	11.1	11.1	103	103	50-150	0	20	
Cadmium	mg/L	ND	10	10	10.1	10.0	101	100	50-150	1	20 F5	
Chromium	mg/L	0.35	10	10	10.7	10.6	103	102	50-150	1	20	
Lead	mg/L	ND	10	10	9.3	9.2	93	92	50-150	1	20	
Selenium	mg/L	ND	10	10	10.6	10.4	106	104	50-150	1	20	
Silver	mg/L	ND	5	5	5.2	5.2	104	103	50-150	1	20	
Zinc	mg/L	ND	10	10	10.1	10	101	99	50-150	1	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

MATRIX SPIKE SAMPLE:		3464281					
Parameter	Units	50354939001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.25	50	57.5	115	50-150	
Barium	mg/L	<1.2	50	53.9	108	50-150	
Cadmium	mg/L	<0.12	50	55.5	111	50-150	F5
Chromium	mg/L	1810	50	1870	121	50-150	
Lead	mg/L	0.62	50	54.0	107	50-150	
Selenium	mg/L	<0.25	50	55.7	111	50-150	
Silver	mg/L	<0.25	25	28.7	115	50-150	
Zinc	mg/L	1260	50	1410	289	50-150	P6

MATRIX SPIKE SAMPLE:		3464282					
Parameter	Units	92690877008 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	178 ug/L	10	11.5	113	50-150	
Barium	mg/L	ND	10	10.7	105	50-150	
Cadmium	mg/L	ND	10	10.4	104	50-150	F5
Chromium	mg/L	ND	10	10.4	104	50-150	
Lead	mg/L	ND	10	9.4	94	50-150	
Selenium	mg/L	2610 ug/L	10	13.3	107	50-150	
Silver	mg/L	ND	5	5.3	105	50-150	
Zinc	mg/L		10	11.0	107	50-150	

MATRIX SPIKE SAMPLE:		3464283					
Parameter	Units	50355005001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.050	10	11.1	111	50-150	
Barium	mg/L	<0.25	10	10.8	108	50-150	
Cadmium	mg/L	<0.025	10	10.4	103	50-150	F5
Chromium	mg/L	0.14	10	10.8	107	50-150	
Lead	mg/L	<0.050	10	9.8	98	50-150	
Selenium	mg/L	<0.050	10	10.8	108	50-150	
Silver	mg/L	<0.050	5	5.3	107	50-150	
Zinc	mg/L	1.1	10	11.7	106	50-150	

MATRIX SPIKE SAMPLE:		3464284					
Parameter	Units	92690842001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	11.2	112	50-150	
Barium	mg/L	ND	10	10.8	108	50-150	
Cadmium	mg/L	ND	10	10.4	104	50-150	F5
Chromium	mg/L	9.8	10	20.5	107	50-150	
Lead	mg/L	ND	10	9.8	98	50-150	
Selenium	mg/L	ND	10	10.9	109	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50355005

MATRIX SPIKE SAMPLE:		3464284					
Parameter	Units	92690842001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Silver	mg/L	ND	5	5.3	107	50-150	
Zinc	mg/L	13000 ug/L	10	23.6	106	50-150	

MATRIX SPIKE SAMPLE:		3464285					
Parameter	Units	92690844001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	11.2	112	50-150	
Barium	mg/L	ND	10	11.0	110	50-150	
Cadmium	mg/L	ND	10	10.5	105	50-150	F5
Chromium	mg/L	3.6	10	14.7	111	50-150	
Lead	mg/L	ND	10	10	100	50-150	
Selenium	mg/L	ND	10	10.9	109	50-150	
Silver	mg/L	ND	5	5.4	108	50-150	
Zinc	mg/L	ND	10	10.9	108	50-150	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 755713

Analysis Method: EPA 6020

QC Batch Method: EPA 3050B

Analysis Description: 6020 MET

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3463305

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Arsenic	mg/kg	<0.0095	0.10	0.0095	10/11/23 16:16	
Cadmium	mg/kg	<0.023	0.050	0.023	10/11/23 16:16	
Chromium	mg/kg	0.038J	0.20	0.025	10/11/23 16:16	
Copper	mg/kg	<0.041	0.10	0.041	10/11/23 16:16	
Lead	mg/kg	<0.0039	0.10	0.0039	10/11/23 16:16	
Molybdenum	mg/kg	<0.0092	0.10	0.0092	10/11/23 16:16	N2
Nickel	mg/kg	0.015J	0.050	0.013	10/11/23 16:16	
Selenium	mg/kg	<0.028	0.10	0.028	10/11/23 16:16	
Silver	mg/kg	<0.0022	0.050	0.0022	10/11/23 16:16	
Zinc	mg/kg	0.15J	0.50	0.063	10/11/23 16:16	

LABORATORY CONTROL SAMPLE: 3463306

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.9	99	80-120	
Cadmium	mg/kg	4	4.0	99	80-120	
Chromium	mg/kg	4	4.0	101	80-120	
Copper	mg/kg	4	4.0	99	80-120	
Lead	mg/kg	4	4.0	100	80-120	
Molybdenum	mg/kg	4	4.1	103	80-120	N2
Nickel	mg/kg	4	4.2	105	80-120	
Selenium	mg/kg	4	4.0	99	80-120	
Silver	mg/kg	4	4.1	102	80-120	
Zinc	mg/kg	4	4.0	99	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3463307

3463308

Parameter	Units	52124347001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	ND	7.9	7.9	8.3	8.2	103	103	75-125	0	20	
Cadmium	mg/kg	ND	7.9	7.9	7.9	8.0	99	100	75-125	2	20	
Chromium	mg/kg	ND	7.9	7.9	8.5	8.5	105	104	75-125	1	20	
Copper	mg/kg	ND	7.9	7.9	8.1	8.2	100	101	75-125	1	20	
Lead	mg/kg	ND	7.9	7.9	8.0	8.1	100	101	75-125	1	20	
Molybdenum	mg/kg	ND	7.9	7.9	8.4	8.3	105	104	75-125	1	20	N2
Nickel	mg/kg	0.22	7.9	7.9	8.2	8.2	100	100	75-125	0	20	
Selenium	mg/kg	ND	7.9	7.9	8.8	9.0	110	113	75-125	2	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3463307 3463308												
Parameter	Units	52124347001 Result	MS	MSD	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max	
			Spike Conc.	Spike Conc.							RPD	Qual
Silver	mg/kg	ND	7.9	7.9	8.1	8.2	101	103	75-125	2	20	
Zinc	mg/kg	ND	7.9	7.9	8.1	8.2	96	98	75-125	1	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 755799

Analysis Method: EPA 8260

QC Batch Method: EPA 8260

Analysis Description: 8260 MSV 5030 Low

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3463668

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	<0.00035	0.0050	0.00035	10/04/23 14:35	
1,1,1-Trichloroethane	mg/kg	<0.00034	0.0050	0.00034	10/04/23 14:35	
1,1,2,2-Tetrachloroethane	mg/kg	<0.00037	0.0050	0.00037	10/04/23 14:35	
1,1,2-Trichloroethane	mg/kg	<0.00037	0.0050	0.00037	10/04/23 14:35	
1,1-Dichloroethane	mg/kg	<0.00034	0.0050	0.00034	10/04/23 14:35	
1,1-Dichloroethene	mg/kg	<0.00035	0.0050	0.00035	10/04/23 14:35	
1,2,3-Trichloropropane	mg/kg	<0.00038	0.0050	0.00038	10/04/23 14:35	
1,2-Dibromo-3-chloropropane	mg/kg	<0.0011	0.010	0.0011	10/04/23 14:35	
1,2-Dibromoethane (EDB)	mg/kg	<0.00036	0.0050	0.00036	10/04/23 14:35	
1,2-Dichlorobenzene	mg/kg	<0.00040	0.0050	0.00040	10/04/23 14:35	
1,2-Dichloroethane	mg/kg	<0.00033	0.0050	0.00033	10/04/23 14:35	
1,2-Dichloropropane	mg/kg	<0.00036	0.0050	0.00036	10/04/23 14:35	
1,3-Dichlorobenzene	mg/kg	<0.00044	0.0050	0.00044	10/04/23 14:35	
1,4-Dichlorobenzene	mg/kg	<0.00044	0.0050	0.00044	10/04/23 14:35	
2-Butanone (MEK)	mg/kg	<0.0034	0.025	0.0034	10/04/23 14:35	
2-Hexanone	mg/kg	<0.0022	0.10	0.0022	10/04/23 14:35	
4-Methyl-2-pentanone (MIBK)	mg/kg	<0.0021	0.025	0.0021	10/04/23 14:35	
Acetone	mg/kg	<0.0060	0.10	0.0060	10/04/23 14:35	
Acrolein	mg/kg	<0.018	0.10	0.018	10/04/23 14:35	
Acrylonitrile	mg/kg	<0.0020	0.10	0.0020	10/04/23 14:35	
Benzene	mg/kg	<0.00049	0.0050	0.00049	10/04/23 14:35	
Bromodichloromethane	mg/kg	<0.00032	0.0050	0.00032	10/04/23 14:35	
Bromoform	mg/kg	<0.00031	0.0050	0.00031	10/04/23 14:35	
Bromomethane	mg/kg	<0.0017	0.0050	0.0017	10/04/23 14:35	
Carbon disulfide	mg/kg	<0.00039	0.010	0.00039	10/04/23 14:35	
Carbon tetrachloride	mg/kg	<0.0016	0.0050	0.0016	10/04/23 14:35	
Chlorobenzene	mg/kg	<0.00036	0.0050	0.00036	10/04/23 14:35	
Chloroethane	mg/kg	<0.00099	0.0050	0.00099	10/04/23 14:35	
Chloroform	mg/kg	<0.0032	0.0050	0.0032	10/04/23 14:35	
Chloromethane	mg/kg	<0.00047	0.0050	0.00047	10/04/23 14:35	
cis-1,2-Dichloroethene	mg/kg	<0.00037	0.0050	0.00037	10/04/23 14:35	
cis-1,3-Dichloropropene	mg/kg	<0.00035	0.0050	0.00035	10/04/23 14:35	
Dibromochloromethane	mg/kg	<0.00029	0.0050	0.00029	10/04/23 14:35	
Dibromomethane	mg/kg	<0.00036	0.0050	0.00036	10/04/23 14:35	
Dichlorodifluoromethane	mg/kg	<0.00033	0.0050	0.00033	10/04/23 14:35	
Ethyl methacrylate	mg/kg	<0.00038	0.10	0.00038	10/04/23 14:35	
Ethylbenzene	mg/kg	<0.0010	0.0050	0.0010	10/04/23 14:35	
Iodomethane	mg/kg	<0.0016	0.10	0.0016	10/04/23 14:35	
Methyl methacrylate	mg/kg	<0.00051	0.10	0.00051	10/04/23 14:35	
Methyl-tert-butyl ether	mg/kg	<0.00034	0.0050	0.00034	10/04/23 14:35	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

METHOD BLANK: 3463668

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Methylene Chloride	mg/kg	<0.014	0.020	0.014	10/04/23 14:35	
Styrene	mg/kg	<0.00050	0.0050	0.00050	10/04/23 14:35	
Tetrachloroethene	mg/kg	<0.00037	0.0050	0.00037	10/04/23 14:35	
Toluene	mg/kg	<0.00046	0.0050	0.00046	10/04/23 14:35	
trans-1,2-Dichloroethene	mg/kg	<0.00038	0.0050	0.00038	10/04/23 14:35	
trans-1,3-Dichloropropene	mg/kg	<0.00034	0.0050	0.00034	10/04/23 14:35	
trans-1,4-Dichloro-2-butene	mg/kg	<0.00055	0.10	0.00055	10/04/23 14:35	
Trichloroethene	mg/kg	<0.00044	0.0050	0.00044	10/04/23 14:35	
Trichlorofluoromethane	mg/kg	<0.00034	0.0050	0.00034	10/04/23 14:35	
Vinyl acetate	mg/kg	<0.0019	0.10	0.0019	10/04/23 14:35	
Vinyl chloride	mg/kg	<0.00037	0.0050	0.00037	10/04/23 14:35	
Xylene (Total)	mg/kg	<0.0038	0.010	0.0038	10/04/23 14:35	
4-Bromofluorobenzene (S)	%	103	63-132		10/04/23 14:35	
Dibromofluoromethane (S)	%	103	75-135		10/04/23 14:35	
Toluene-d8 (S)	%	97	65-148		10/04/23 14:35	

LABORATORY CONTROL SAMPLE: 3463669

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	0.05	0.049	98	74-134	
1,1,1-Trichloroethane	mg/kg	0.05	0.053	106	66-133	
1,1,2,2-Tetrachloroethane	mg/kg	0.05	0.047	95	62-131	
1,1,2-Trichloroethane	mg/kg	0.05	0.047	95	73-128	
1,1-Dichloroethane	mg/kg	0.05	0.055	110	67-128	
1,1-Dichloroethene	mg/kg	0.05	0.059	118	65-138	
1,2,3-Trichloropropane	mg/kg	0.05	0.051	101	69-128	
1,2-Dibromo-3-chloropropane	mg/kg	0.05	0.046	92	62-133	
1,2-Dibromoethane (EDB)	mg/kg	0.05	0.052	104	74-131	
1,2-Dichlorobenzene	mg/kg	0.05	0.047	94	68-130	
1,2-Dichloroethane	mg/kg	0.05	0.059	119	62-129	
1,2-Dichloropropane	mg/kg	0.05	0.050	101	64-132	
1,3-Dichlorobenzene	mg/kg	0.05	0.047	94	66-131	
1,4-Dichlorobenzene	mg/kg	0.05	0.048	95	66-129	
2-Butanone (MEK)	mg/kg	0.25	0.27	108	57-139	
2-Hexanone	mg/kg	0.25	0.28	111	59-130	
4-Methyl-2-pentanone (MIBK)	mg/kg	0.25	0.31	122	62-132	
Acetone	mg/kg	0.25	0.29	115	45-139	
Acrolein	mg/kg	1	1.6	158	59-123 L1	
Acrylonitrile	mg/kg	0.25	0.28	111	64-135	
Benzene	mg/kg	0.05	0.051	103	65-128	
Bromodichloromethane	mg/kg	0.05	0.053	105	69-132	
Bromoform	mg/kg	0.05	0.046	92	69-140	
Bromomethane	mg/kg	0.05	0.079	158	42-162	
Carbon disulfide	mg/kg	0.05	0.059	118	64-132	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

LABORATORY CONTROL SAMPLE: 3463669

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Carbon tetrachloride	mg/kg	0.05	0.054	108	65-141	
Chlorobenzene	mg/kg	0.05	0.049	99	69-124	
Chloroethane	mg/kg	0.05	0.071	141	64-136	L1
Chloroform	mg/kg	0.05	0.055	109	67-122	
Chloromethane	mg/kg	0.05	0.070	139	49-130	L1
cis-1,2-Dichloroethene	mg/kg	0.05	0.049	99	64-131	
cis-1,3-Dichloropropene	mg/kg	0.05	0.052	103	72-134	
Dibromochloromethane	mg/kg	0.05	0.050	100	73-136	
Dibromomethane	mg/kg	0.05	0.054	108	70-131	
Dichlorodifluoromethane	mg/kg	0.05	0.052	105	10-117	
Ethyl methacrylate	mg/kg	0.05	0.051J	101	70-132	
Ethylbenzene	mg/kg	0.05	0.050	99	67-127	
Iodomethane	mg/kg	0.05	0.056J	113	42-150	
Methyl methacrylate	mg/kg	0.05	0.046J	92	62-140	
Methyl-tert-butyl ether	mg/kg	0.05	0.051	101	66-135	
Methylene Chloride	mg/kg	0.05	0.058	115	66-137	
Styrene	mg/kg	0.05	0.050	99	72-129	
Tetrachloroethene	mg/kg	0.05	0.049	99	62-135	
Toluene	mg/kg	0.05	0.050	100	65-123	
trans-1,2-Dichloroethene	mg/kg	0.05	0.050	100	63-131	
trans-1,3-Dichloropropene	mg/kg	0.05	0.051	102	71-135	
trans-1,4-Dichloro-2-butene	mg/kg	0.05	0.059J	118	62-156	
Trichloroethene	mg/kg	0.05	0.050	101	64-135	
Trichlorofluoromethane	mg/kg	0.05	0.068	137	56-142	
Vinyl acetate	mg/kg	0.2	0.28	141	47-142	
Vinyl chloride	mg/kg	0.05	0.062	125	54-134	
Xylene (Total)	mg/kg	0.15	0.15	98	65-124	
4-Bromofluorobenzene (S)	%			102	63-132	
Dibromofluoromethane (S)	%			102	75-135	
Toluene-d8 (S)	%			98	65-148	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3463670 3463671

Parameter	Units	50355363001		MS	MSD	MS	MSD	MS	MSD	% Rec	RPD	Max	Qual
		Result	Spike Conc.	Spike Conc.	Result								
1,1,1,2-Tetrachloroethane	mg/kg	ND	0.046	0.047	0.042	0.043	92	92	26-163	2	20		
1,1,1-Trichloroethane	mg/kg	ND	0.046	0.047	0.049	0.051	107	108	61-143	3	20		
1,1,2,2-Tetrachloroethane	mg/kg	ND	0.046	0.047	<0.00034	0.0011J	0	2	30-172		20	M1	
1,1,2-Trichloroethane	mg/kg	ND	0.046	0.047	0.022	0.017	47	37	27-162	23	20	R1	
1,1-Dichloroethane	mg/kg	ND	0.046	0.047	0.051	0.052	110	111	52-145	3	20		
1,1-Dichloroethene	mg/kg	ND	0.046	0.047	0.079	0.086	171	183	58-154	10	20	M1	
1,2,3-Trichloropropane	mg/kg	ND	0.046	0.047	0.046	0.049	100	103	20-178	5	20		
1,2-Dibromo-3-chloropropane	mg/kg	ND	0.046	0.047	0.020	0.015	44	32	10-161	27	20	R1	
1,2-Dibromoethane (EDB)	mg/kg	ND	0.046	0.047	0.048	0.051	104	107	42-153	5	20		

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3463670				3463671								
		50355363001	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Parameter	Units	Result										
1,2-Dichlorobenzene	mg/kg	ND	0.046	0.047	0.047	0.048	102	102	10-164	3	20	
1,2-Dichloroethane	mg/kg	ND	0.046	0.047	0.055	0.057	120	121	58-135	3	20	
1,2-Dichloropropane	mg/kg	ND	0.046	0.047	0.048	0.050	104	106	48-150	4	20	
1,3-Dichlorobenzene	mg/kg	ND	0.046	0.047	0.047	0.048	102	102	10-163	3	20	
1,4-Dichlorobenzene	mg/kg	ND	0.046	0.047	0.044	0.044	95	94	10-160	1	20	
2-Butanone (MEK)	mg/kg	0.026	0.23	0.24	0.27	0.29	105	114	28-183	9	20	
2-Hexanone	mg/kg	ND	0.23	0.24	0.24	0.26	105	112	26-159	9	20	
4-Methyl-2-pentanone (MIBK)	mg/kg	ND	0.23	0.24	0.27	0.30	116	125	24-167	10	20	
Acetone	mg/kg	0.19	0.23	0.24	0.43	0.47	105	121	10-194	10	20	
Acrolein	mg/kg	ND	0.92	0.95	0.077J	0.079J	8	8	10-155		20	M0
Acrylonitrile	mg/kg	ND	0.23	0.24	0.22	0.24	96	100	32-169	6	20	
Benzene	mg/kg	ND	0.046	0.047	0.046	0.048	101	102	42-152	4	20	
Bromodichloromethane	mg/kg	ND	0.046	0.047	0.040	0.040	87	84	26-164	1	20	
Bromoform	mg/kg	ND	0.046	0.047	0.044	0.045	96	95	14-177	2	20	
Bromomethane	mg/kg	ND	0.046	0.047	0.074	0.073	161	154	21-164	2	20	
Carbon disulfide	mg/kg	ND	0.046	0.047	0.043	0.046	93	98	44-143	9	20	
Carbon tetrachloride	mg/kg	ND	0.046	0.047	0.051	0.052	110	110	32-170	3	20	
Chlorobenzene	mg/kg	ND	0.046	0.047	0.044	0.046	95	97	27-149	5	20	
Chloroethane	mg/kg	ND	0.046	0.047	0.075	0.078	163	166	42-158	4	20	M0
Chloroform	mg/kg	ND	0.046	0.047	0.050	0.052	110	110	54-143	3	20	
Chloromethane	mg/kg	ND	0.046	0.047	0.062	0.063	135	134	39-143	2	20	
cis-1,2-Dichloroethene	mg/kg	ND	0.046	0.047	0.046	0.048	101	102	56-140	4	20	
cis-1,3-Dichloropropene	mg/kg	ND	0.046	0.047	0.045	0.046	98	98	31-155	3	20	
Dibromochloromethane	mg/kg	ND	0.046	0.047	0.041	0.041	89	87	23-164	0	20	
Dibromomethane	mg/kg	ND	0.046	0.047	0.049	0.053	106	112	48-147	8	20	
Dichlorodifluoromethane	mg/kg	ND	0.046	0.047	0.052	0.052	112	111	10-121	2	20	
Ethyl methacrylate	mg/kg	ND	0.046	0.047	<0.0003 5	0.0014J 8	0	3	10-186		20	M1
Ethylbenzene	mg/kg	ND	0.046	0.047	0.047	0.049	101	103	27-150	4	20	
Iodomethane	mg/kg	ND	0.046	0.047	0.043J	0.050J	94	105	23-162		20	
Methyl methacrylate	mg/kg	ND	0.046	0.047	<0.0004 7	<0.0004 8	0	0	10-200		20	M1
Methyl-tert-butyl ether	mg/kg	ND	0.046	0.047	0.046	0.050	101	106	68-143	8	20	
Methylene Chloride	mg/kg	ND	0.046	0.047	0.061	0.065	117	123	43-151	7	20	
Styrene	mg/kg	ND	0.046	0.047	0.043	0.045	94	96	10-186	4	20	
Tetrachloroethene	mg/kg	ND	0.046	0.047	0.045	0.048	98	102	40-157	6	20	
Toluene	mg/kg	ND	0.046	0.047	0.045	0.048	98	100	32-149	5	20	
trans-1,2-Dichloroethene	mg/kg	ND	0.046	0.047	0.047	0.048	102	102	51-146	3	20	
trans-1,3-Dichloropropene	mg/kg	ND	0.046	0.047	0.047	0.049	102	103	14-169	4	20	
trans-1,4-Dichloro-2-butene	mg/kg	ND	0.046	0.047	0.051J	0.052J	110	111	10-182		20	
Trichloroethene	mg/kg	ND	0.046	0.047	0.087	0.091	188	192	34-168	5	20	M1
Trichlorofluoromethane	mg/kg	ND	0.046	0.047	0.065	0.067	142	141	35-158	2	20	
Vinyl acetate	mg/kg	ND	0.18	0.19	0.048J	0.051J	26	27	10-156		20	
Vinyl chloride	mg/kg	ND	0.046	0.047	0.059	0.060	128	126	49-148	1	20	
Xylene (Total)	mg/kg	ND	0.14	0.14	0.14	0.14	100	102	23-151	4	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3463670 3463671												
Parameter	Units	50355363001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
4-Bromofluorobenzene (S)	%.						99	100	63-132			
Dibromofluoromethane (S)	%.						17	16	75-135			S0
Toluene-d8 (S)	%.						100	100	65-148			

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 757059

Analysis Method: EPA 5030/8260

QC Batch Method: EPA 5030/8260

Analysis Description: 8260 MSV TCLP

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3469614

Matrix: Water

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	<0.025	0.050	0.025	10/12/23 14:11	
1,2-Dichloroethane	mg/L	<0.025	0.050	0.025	10/12/23 14:11	
2-Butanone (MEK)	mg/L	<0.50	1.0	0.50	10/12/23 14:11	
Benzene	mg/L	<0.010	0.050	0.010	10/12/23 14:11	
Carbon tetrachloride	mg/L	<0.025	0.050	0.025	10/12/23 14:11	
Chlorobenzene	mg/L	<0.025	0.050	0.025	10/12/23 14:11	
Chloroform	mg/L	<0.025	0.050	0.025	10/12/23 14:11	
Tetrachloroethene	mg/L	<0.025	0.050	0.025	10/12/23 14:11	
Trichloroethene	mg/L	<0.025	0.050	0.025	10/12/23 14:11	
Vinyl chloride	mg/L	<0.010	0.020	0.010	10/12/23 14:11	
4-Bromofluorobenzene (S)	%	101	79-124		10/12/23 14:11	
Dibromofluoromethane (S)	%	109	82-128		10/12/23 14:11	
Toluene-d8 (S)	%	99	73-122		10/12/23 14:11	

LABORATORY CONTROL SAMPLE: 3469615

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	0.5	0.58	117	73-133	
1,2-Dichloroethane	mg/L	0.5	0.57	113	70-124	
2-Butanone (MEK)	mg/L	2.5	2.8	114	59-134	
Benzene	mg/L	0.5	0.52	104	74-124	
Carbon tetrachloride	mg/L	0.5	0.53	107	78-132	
Chlorobenzene	mg/L	0.5	0.50	100	77-121	
Chloroform	mg/L	0.5	0.54	108	75-118	
Tetrachloroethene	mg/L	0.5	0.51	102	73-132	
Trichloroethene	mg/L	0.5	0.50	101	75-127	
Vinyl chloride	mg/L	0.5	0.62	124	48-133	
4-Bromofluorobenzene (S)	%			99	79-124	
Dibromofluoromethane (S)	%			104	82-128	
Toluene-d8 (S)	%			100	73-122	

MATRIX SPIKE SAMPLE: 3469616

Parameter	Units	50355349001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.48	96	65-139	
1,2-Dichloroethane	mg/L	ND	0.5	0.48	97	55-146	
2-Butanone (MEK)	mg/L	ND	2.5	2.1	84	45-155	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

MATRIX SPIKE SAMPLE:		3469616					
Parameter	Units	50355349001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Benzene	mg/L	ND	0.5	0.44	88	65-137	
Carbon tetrachloride	mg/L	ND	0.5	0.45	89	65-156	
Chlorobenzene	mg/L	ND	0.5	0.43	87	54-135	
Chloroform	mg/L	ND	0.5	0.45	91	64-133	
Tetrachloroethene	mg/L	ND	0.5	0.44	87	43-149	
Trichloroethene	mg/L	ND	0.5	0.43	85	52-145	
Vinyl chloride	mg/L	ND	0.5	0.51	102	43-139	
4-Bromofluorobenzene (S)	%				100	79-124	
Dibromofluoromethane (S)	%				104	82-128	
Toluene-d8 (S)	%				101	73-122	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 756328

Analysis Method: EPA 8081

QC Batch Method: EPA 3546

Analysis Description: 8081 GCS Pesticides

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3466432

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<0.85	5.0	0.85	10/09/23 19:04	
4,4'-DDE	ug/kg	<0.99	5.0	0.99	10/09/23 19:04	
4,4'-DDT	ug/kg	<2.6	5.0	2.6	10/09/23 19:04	
Aldrin	ug/kg	<0.37	2.5	0.37	10/09/23 19:04	
alpha-BHC	ug/kg	<0.36	2.5	0.36	10/09/23 19:04	
alpha-Chlordane	ug/kg	<0.94	2.5	0.94	10/09/23 19:04	
beta-BHC	ug/kg	<0.47	2.5	0.47	10/09/23 19:04	
Chlordane (Technical)	ug/kg	<23.2	50.0	23.2	10/09/23 19:04	
delta-BHC	ug/kg	<0.47	2.5	0.47	10/09/23 19:04	
Dieldrin	ug/kg	<0.86	5.0	0.86	10/09/23 19:04	
Endosulfan I	ug/kg	<0.44	2.5	0.44	10/09/23 19:04	
Endosulfan II	ug/kg	<1.8	5.0	1.8	10/09/23 19:04	
Endosulfan sulfate	ug/kg	<0.94	5.0	0.94	10/09/23 19:04	
Endrin	ug/kg	<0.79	5.0	0.79	10/09/23 19:04	
Endrin aldehyde	ug/kg	<1.5	5.0	1.5	10/09/23 19:04	
gamma-BHC (Lindane)	ug/kg	<0.38	2.5	0.38	10/09/23 19:04	
gamma-Chlordane	ug/kg	<1.3	2.5	1.3	10/09/23 19:04	
Heptachlor	ug/kg	<0.48	2.5	0.48	10/09/23 19:04	
Heptachlor epoxide	ug/kg	<0.41	2.5	0.41	10/09/23 19:04	
Toxaphene	ug/kg	<14.5	50.0	14.5	10/09/23 19:04	
Decachlorobiphenyl (S)	%	86	10-166		10/09/23 19:04	

LABORATORY CONTROL SAMPLE: 3466433

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	20	19.5	97	45-171	
4,4'-DDE	ug/kg	20	20.1	100	45-163	
4,4'-DDT	ug/kg	20	21.8	109	33-183	
Aldrin	ug/kg	10	10.3	103	50-151	
alpha-BHC	ug/kg	10	9.6	96	49-148	
alpha-Chlordane	ug/kg	10	9.9	99	46-150	
beta-BHC	ug/kg	10	10.3	103	51-158	
delta-BHC	ug/kg	10	7.0	70	25-154	
Dieldrin	ug/kg	20	21.5	108	48-164	
Endosulfan I	ug/kg	10	10.6	106	45-160	
Endosulfan II	ug/kg	20	20.0	100	49-163	
Endosulfan sulfate	ug/kg	20	18.0	90	41-154	
Endrin	ug/kg	20	21.2	106	48-163	
Endrin aldehyde	ug/kg	20	23.9	119	32-180	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

LABORATORY CONTROL SAMPLE: 3466433

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/kg	10	10.0	100	51-152	
gamma-Chlordane	ug/kg	10	11.1	111	44-173	
Heptachlor	ug/kg	10	10.1	101	52-155	
Heptachlor epoxide	ug/kg	10	10.1	101	47-154	
Decachlorobiphenyl (S)	%.			97	10-166	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3466434 3466435

Parameter	Units	5035504001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
4,4'-DDD	ug/kg	ND	25.5	25.4	24.4	24.6	95	97	10-145	1	20	
4,4'-DDE	ug/kg	ND	25.5	25.4	23.8	23.4	93	92	10-136	2	20	
4,4'-DDT	ug/kg	ND	25.5	25.4	27.1	27.4	106	108	10-160	1	20	
Aldrin	ug/kg	ND	12.8	12.8	12.9	12.6	101	99	10-155	2	20	
alpha-BHC	ug/kg	ND	12.8	12.8	12.7	13.5	100	106	10-151	6	20	
alpha-Chlordane	ug/kg	ND	12.8	12.8	12.7	12.7	99	100	10-148	0	20	
beta-BHC	ug/kg	ND	12.8	12.8	12.5	13.0	98	102	10-160	4	20	
delta-BHC	ug/kg	ND	12.8	12.8	8.9	9.4	70	74	10-122	5	20	
Dieldrin	ug/kg	ND	25.5	25.4	26.1	26.1	102	103	10-182	0	20	
Endosulfan I	ug/kg	ND	12.8	12.8	13.0	12.4	102	98	10-145	4	20	
Endosulfan II	ug/kg	ND	25.5	25.4	24.1	24.2	94	95	10-168	1	20	
Endosulfan sulfate	ug/kg	ND	25.5	25.4	21.9	21.5	86	85	13-124	2	20	
Endrin	ug/kg	ND	25.5	25.4	24.9	25.4	97	100	10-148	2	20	
Endrin aldehyde	ug/kg	ND	25.5	25.4	28.8	29.0	113	114	10-169	0	20	
gamma-BHC (Lindane)	ug/kg	ND	12.8	12.8	13.3	13.9	104	109	10-153	5	20	
gamma-Chlordane	ug/kg	ND	12.8	12.8	15.3	<1.7	120	0	10-173		20	M1
Heptachlor	ug/kg	ND	12.8	12.8	13.5	14.1	106	111	10-158	4	20	
Heptachlor epoxide	ug/kg	ND	12.8	12.8	13.3	12.5	104	98	10-172	6	20	
Decachlorobiphenyl (S)	%.						54	56	10-166			

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 756371

Analysis Method: EPA 8081

QC Batch Method: EPA 3510

Analysis Description: 8081 GCS TCLP Pesticides

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3466566

Matrix: Water

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Chlordane (Technical)	mg/L	<0.0010	0.0010	0.0010	10/10/23 15:23	
Endrin	mg/L	<0.00010	0.00010	0.00010	10/10/23 15:23	
gamma-BHC (Lindane)	mg/L	<0.000050	0.000050	0.000050	10/10/23 15:23	
Heptachlor	mg/L	<0.000050	0.000050	0.000050	10/10/23 15:23	
Heptachlor epoxide	mg/L	<0.000050	0.000050	0.000050	10/10/23 15:23	
Methoxychlor	mg/L	<0.00050	0.00050	0.00050	10/10/23 15:23	
Toxaphene	mg/L	<0.0010	0.0010	0.0010	10/10/23 15:23	
Decachlorobiphenyl (S)	%.	82	10-124		10/10/23 15:23	1d

LABORATORY CONTROL SAMPLE: 3466567

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	0.002	0.0017	85	29-177	
gamma-BHC (Lindane)	mg/L	0.001	0.00077	77	26-163	
Heptachlor	mg/L	0.001	0.00045	45	12-157	
Heptachlor epoxide	mg/L	0.001	0.00076	76	27-168	
Methoxychlor	mg/L	0.01	0.0081	81	28-174	
Decachlorobiphenyl (S)	%.			60	10-124	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 755208

QC Batch Method: EPA 3546

Analysis Method: EPA 8082

Analysis Description: 8082 PCB Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3461452

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<3.7	100	3.7	10/02/23 19:48	
PCB-1221 (Aroclor 1221)	ug/kg	<4.7	100	4.7	10/02/23 19:48	
PCB-1232 (Aroclor 1232)	ug/kg	<5.2	100	5.2	10/02/23 19:48	
PCB-1242 (Aroclor 1242)	ug/kg	<4.4	100	4.4	10/02/23 19:48	
PCB-1248 (Aroclor 1248)	ug/kg	<3.1	100	3.1	10/02/23 19:48	
PCB-1254 (Aroclor 1254)	ug/kg	<4.5	100	4.5	10/02/23 19:48	
PCB-1260 (Aroclor 1260)	ug/kg	<4.7	100	4.7	10/02/23 19:48	
Tetrachloro-m-xylene (S)	%	81	10-133		10/02/23 19:48	

LABORATORY CONTROL SAMPLE: 3461453

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	333	296	89	50-120	
PCB-1260 (Aroclor 1260)	ug/kg	333	296	89	40-122	
Tetrachloro-m-xylene (S)	%			87	10-133	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3461454 3461455

Parameter	Units	50355009001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/kg	ND	403	391	286	288	71	74	10-154	1	20	
PCB-1260 (Aroclor 1260)	ug/kg	ND	403	391	255	263	63	67	10-165	3	20	
Tetrachloro-m-xylene (S)	%						67	59	10-133			

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 756345

QC Batch Method: EPA 8151

Analysis Method: EPA 8151A

Analysis Description: 8151 GCS TCLP Herbicides

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3466483

Matrix: Water

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0010	0.0010	0.0010	10/10/23 22:58	
2,4-D	mg/L	<0.0010	0.0010	0.0010	10/10/23 22:58	
2,4-DCAA (S)	%.	79	22-132		10/10/23 22:58	

LABORATORY CONTROL SAMPLE: 3466484

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	0.005	0.0032	65	37-122	
2,4-D	mg/L	0.005	0.0036	71	24-129	
2,4-DCAA (S)	%.			68	22-132	

MATRIX SPIKE SAMPLE: 3466485

Parameter	Units	50355005001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0050	0.025	0.018	74	20-155	
2,4-D	mg/L	<0.0050	0.025	0.020	79	12-159	
2,4-DCAA (S)	%.				89	22-132	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 755461

Analysis Method: EPA 8270

QC Batch Method: EPA 3546

Analysis Description: 8270 Solid MSSV Microwave Short Spike

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3462213

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
1-Methylnaphthalene	ug/kg	<121	330	121	10/04/23 22:00	
2,4,5-Trichlorophenol	ug/kg	<115	330	115	10/04/23 22:00	
2,4,6-Trichlorophenol	ug/kg	<101	330	101	10/04/23 22:00	
2,4-Dichlorophenol	ug/kg	<114	330	114	10/04/23 22:00	
2,4-Dimethylphenol	ug/kg	<114	330	114	10/04/23 22:00	
2,4-Dinitrophenol	ug/kg	<182	1600	182	10/04/23 22:00	
2,4-Dinitrotoluene	ug/kg	<110	330	110	10/04/23 22:00	
2,6-Dinitrotoluene	ug/kg	<94.3	330	94.3	10/04/23 22:00	
2-Chloronaphthalene	ug/kg	<92.7	330	92.7	10/04/23 22:00	
2-Chlorophenol	ug/kg	<118	330	118	10/04/23 22:00	
2-Methylnaphthalene	ug/kg	<98.5	330	98.5	10/04/23 22:00	
2-Methylphenol(o-Cresol)	ug/kg	<141	330	141	10/04/23 22:00	
2-Nitroaniline	ug/kg	<135	330	135	10/04/23 22:00	
2-Nitrophenol	ug/kg	<128	330	128	10/04/23 22:00	
3&4-Methylphenol(m&p Cresol)	ug/kg	<138	660	138	10/04/23 22:00	
3,3'-Dichlorobenzidine	ug/kg	<327	660	327	10/04/23 22:00	
3-Nitroaniline	ug/kg	<119	330	119	10/04/23 22:00	
4,6-Dinitro-2-methylphenol	ug/kg	<201	660	201	10/04/23 22:00	
4-Bromophenylphenyl ether	ug/kg	<123	330	123	10/04/23 22:00	
4-Chloro-3-methylphenol	ug/kg	<134	660	134	10/04/23 22:00	
4-Chloroaniline	ug/kg	<86.9	660	86.9	10/04/23 22:00	
4-Chlorophenylphenyl ether	ug/kg	<102	330	102	10/04/23 22:00	
4-Nitroaniline	ug/kg	<131	330	131	10/04/23 22:00	
4-Nitrophenol	ug/kg	<250	1600	250	10/04/23 22:00	
Acenaphthene	ug/kg	<88.1	330	88.1	10/04/23 22:00	
Acenaphthylene	ug/kg	<98.7	330	98.7	10/04/23 22:00	
Anthracene	ug/kg	<136	330	136	10/04/23 22:00	
Benzo(a)anthracene	ug/kg	<98.0	330	98.0	10/04/23 22:00	
Benzo(a)pyrene	ug/kg	<108	330	108	10/04/23 22:00	
Benzo(b)fluoranthene	ug/kg	<108	330	108	10/04/23 22:00	
Benzo(g,h,i)perylene	ug/kg	<120	330	120	10/04/23 22:00	
Benzo(k)fluoranthene	ug/kg	<120	330	120	10/04/23 22:00	
Benzyl alcohol	ug/kg	<106	660	106	10/04/23 22:00	
bis(2-Chloroethoxy)methane	ug/kg	<106	330	106	10/04/23 22:00	
bis(2-Chloroethyl) ether	ug/kg	<127	330	127	10/04/23 22:00	
bis(2-Ethylhexyl)phthalate	ug/kg	<101	330	101	10/04/23 22:00	
bis(2chloro1methylethyl) ether	ug/kg	<115	330	115	10/04/23 22:00	
Butylbenzylphthalate	ug/kg	<181	330	181	10/04/23 22:00	
Chrysene	ug/kg	<108	330	108	10/04/23 22:00	
Di-n-butylphthalate	ug/kg	<121	330	121	10/04/23 22:00	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

METHOD BLANK: 3462213

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Di-n-octylphthalate	ug/kg	<121	330	121	10/04/23 22:00	
Dibenz(a,h)anthracene	ug/kg	<118	330	118	10/04/23 22:00	
Dibenzofuran	ug/kg	<103	330	103	10/04/23 22:00	
Diethylphthalate	ug/kg	<278	330	278	10/04/23 22:00	
Dimethylphthalate	ug/kg	<111	330	111	10/04/23 22:00	
Fluoranthene	ug/kg	<126	330	126	10/04/23 22:00	
Fluorene	ug/kg	<111	330	111	10/04/23 22:00	
Hexachloro-1,3-butadiene	ug/kg	<89.6	330	89.6	10/04/23 22:00	
Hexachlorobenzene	ug/kg	<83.5	330	83.5	10/04/23 22:00	
Hexachlorocyclopentadiene	ug/kg	<161	330	161	10/04/23 22:00	
Hexachloroethane	ug/kg	<100	330	100	10/04/23 22:00	
Indeno(1,2,3-cd)pyrene	ug/kg	<120	330	120	10/04/23 22:00	
Isophorone	ug/kg	<107	330	107	10/04/23 22:00	
N-Nitroso-di-n-propylamine	ug/kg	<127	330	127	10/04/23 22:00	
N-Nitrosodiphenylamine	ug/kg	<107	330	107	10/04/23 22:00	
Naphthalene	ug/kg	<94.8	330	94.8	10/04/23 22:00	
Nitrobenzene	ug/kg	<110	330	110	10/04/23 22:00	
Pentachlorophenol	ug/kg	<253	1600	253	10/04/23 22:00	
Phenanthrene	ug/kg	<129	330	129	10/04/23 22:00	
Phenol	ug/kg	<121	330	121	10/04/23 22:00	
Pyrene	ug/kg	<104	330	104	10/04/23 22:00	
2,4,6-Tribromophenol (S)	%	100	10-121		10/04/23 22:00	
2-Fluorobiphenyl (S)	%	82	33-102		10/04/23 22:00	
2-Fluorophenol (S)	%	84	21-109		10/04/23 22:00	
Nitrobenzene-d5 (S)	%	75	32-95		10/04/23 22:00	
p-Terphenyl-d14 (S)	%	93	20-120		10/04/23 22:00	
Phenol-d5 (S)	%	85	27-116		10/04/23 22:00	

LABORATORY CONTROL SAMPLE: 3462214

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4-Dinitrotoluene	ug/kg	1670	1450	87	59-121	
2-Chlorophenol	ug/kg	1670	1390	84	57-102	
2-Methylnaphthalene	ug/kg	1670	1440	86	52-118	
4-Chloro-3-methylphenol	ug/kg	1670	1550	93	58-135	
4-Nitrophenol	ug/kg	1670	1400J	84	49-133	
Acenaphthene	ug/kg	1670	1490	89	63-105	
Acenaphthylene	ug/kg	1670	1490	89	61-104	
Anthracene	ug/kg	1670	1550	93	66-104	
Benzo(a)anthracene	ug/kg	1670	1880	113	69-109 L1	
Benzo(a)pyrene	ug/kg	1670	1890	113	63-105 L1	
Benzo(b)fluoranthene	ug/kg	1670	2030	122	63-115 L1	
Benzo(g,h,i)perylene	ug/kg	1670	1830	110	64-112	
Benzo(k)fluoranthene	ug/kg	1670	1740	104	65-115	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

LABORATORY CONTROL SAMPLE: 3462214

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chrysene	ug/kg	1670	1780	107	68-109	
Dibenz(a,h)anthracene	ug/kg	1670	1770	106	65-112	
Fluoranthene	ug/kg	1670	1740	105	67-116	
Fluorene	ug/kg	1670	1560	93	66-112	
Indeno(1,2,3-cd)pyrene	ug/kg	1670	1850	111	66-113	
N-Nitroso-di-n-propylamine	ug/kg	1670	1230	74	45-112	
Naphthalene	ug/kg	1670	1350	81	58-103	
Pentachlorophenol	ug/kg	1670	1200J	72	31-123	
Phenanthrene	ug/kg	1670	1670	100	68-106	
Phenol	ug/kg	1670	1320	79	49-109	
Pyrene	ug/kg	1670	1800	108	67-110	
2,4,6-Tribromophenol (S)	%.			97	10-121	
2-Fluorobiphenyl (S)	%.			77	33-102	
2-Fluorophenol (S)	%.			75	21-109	
Nitrobenzene-d5 (S)	%.			70	32-95	
p-Terphenyl-d14 (S)	%.			93	20-120	
Phenol-d5 (S)	%.			76	27-116	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3462215 3462216

Parameter	Units	52123864001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
2,4-Dinitrotoluene	ug/kg	ND	44000	44000	31100	31700	71	72	10-132	2	20	
2-Chlorophenol	ug/kg	ND	44000	44000	33600	34400	76	78	11-121	2	20	
2-Methylnaphthalene	ug/kg	ND	44000	44000	33400	33700	76	77	18-130	1	20	
4-Chloro-3-methylphenol	ug/kg	ND	44000	44000	38600	38100	88	87	17-143	1	20	
4-Nitrophenol	ug/kg	ND	44000	44000	33400J	36400J	76	83	10-148		20	
Acenaphthene	ug/kg	ND	44000	44000	33500	33600	76	76	12-129	0	20	
Acenaphthylene	ug/kg	ND	44000	44000	33800	34400	77	78	10-126	2	20	
Anthracene	ug/kg	ND	44000	44000	35400	35900	81	82	10-137	1	20	
Benzo(a)anthracene	ug/kg	ND	44000	44000	37300	39300	85	90	10-140	5	20	
Benzo(a)pyrene	ug/kg	ND	44000	44000	37700	38900	86	89	10-137	3	20	
Benzo(b)fluoranthene	ug/kg	ND	44000	44000	35200	39000	80	89	10-143	10	20	
Benzo(g,h,i)perylene	ug/kg	ND	44000	44000	37300	39000	85	89	10-143	4	20	
Benzo(k)fluoranthene	ug/kg	ND	44000	44000	40300	38100	92	87	10-146	6	20	
Chrysene	ug/kg	ND	44000	44000	37300	37600	85	86	10-143	1	20	
Dibenz(a,h)anthracene	ug/kg	ND	44000	44000	35600	37300	81	85	13-125	4	20	
Fluoranthene	ug/kg	ND	44000	44000	35600	36900	81	84	10-151	4	20	
Fluorene	ug/kg	ND	44000	44000	34200	35200	78	80	10-136	3	20	
Indeno(1,2,3-cd)pyrene	ug/kg	ND	44000	44000	37200	38700	85	88	10-139	4	20	
N-Nitroso-di-n-propylamine	ug/kg	ND	44000	44000	31100	31900	71	73	13-125	3	20	
Naphthalene	ug/kg	ND	44000	44000	31800	32200	72	73	17-123	1	20	
Pentachlorophenol	ug/kg	ND	44000	44000	37000J	37800J	84	86	10-141		20	
Phenanthrene	ug/kg	ND	44000	44000	34900	35600	79	81	10-149	2	20	
Phenol	ug/kg	ND	44000	44000	32700	33600	74	76	16-120	3	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50355005

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3462215 3462216												
Parameter	Units	52123864001 Result	MS	MSD	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
			Spike Conc.	Spike Conc.								
Pyrene	ug/kg	ND	44000	44000	35400	36200	81	82	10-152	2	20	
2,4,6-Tribromophenol (S)	%.						92	93	10-121			
2-Fluorobiphenyl (S)	%.						78	78	33-102			
2-Fluorophenol (S)	%.						75	75	21-109			
Nitrobenzene-d5 (S)	%.						70	70	32-95			
p-Terphenyl-d14 (S)	%.						87	91	20-120			
Phenol-d5 (S)	%.						79	80	27-116			

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 756452

Analysis Method: EPA 8270

QC Batch Method: EPA 3510

Analysis Description: 8270 TCLP MSSV

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3466854

Matrix: Water

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.0050	0.010	0.0050	10/10/23 18:15	
2,4,5-Trichlorophenol	mg/L	<0.0050	0.050	0.0050	10/10/23 18:15	
2,4,6-Trichlorophenol	mg/L	<0.0050	0.010	0.0050	10/10/23 18:15	
2,4-Dinitrotoluene	mg/L	<0.0050	0.010	0.0050	10/10/23 18:15	
2-Methylphenol(o-Cresol)	mg/L	<0.0050	0.010	0.0050	10/10/23 18:15	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.010	0.020	0.010	10/10/23 18:15	
Hexachloro-1,3-butadiene	mg/L	<0.0050	0.010	0.0050	10/10/23 18:15	
Hexachlorobenzene	mg/L	<0.0050	0.010	0.0050	10/10/23 18:15	
Hexachloroethane	mg/L	<0.0050	0.010	0.0050	10/10/23 18:15	
Nitrobenzene	mg/L	<0.0050	0.010	0.0050	10/10/23 18:15	
Pentachlorophenol	mg/L	<0.025	0.050	0.025	10/10/23 18:15	
Pyridine	mg/L	<0.010	0.010	0.010	10/10/23 18:15	
2,4,6-Tribromophenol (S)	%	78	37-127		10/10/23 18:15	
2-Fluorobiphenyl (S)	%	47	36-92		10/10/23 18:15	
2-Fluorophenol (S)	%	37	23-69		10/10/23 18:15	
Nitrobenzene-d5 (S)	%	53	44-107		10/10/23 18:15	
p-Terphenyl-d14 (S)	%	74	57-146		10/10/23 18:15	
Phenol-d5 (S)	%	23	15-47		10/10/23 18:15	

LABORATORY CONTROL SAMPLE: 3466855

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	0.05	0.021	43	23-87	
2,4,5-Trichlorophenol	mg/L	0.05	0.032J	63	43-131	
2,4,6-Trichlorophenol	mg/L	0.05	0.033	66	42-128	
2,4-Dinitrotoluene	mg/L	0.05	0.030	60	47-116	
2-Methylphenol(o-Cresol)	mg/L	0.05	0.029	58	35-98	
3&4-Methylphenol(m&p Cresol)	mg/L	0.1	0.054	54	30-91	
Hexachloro-1,3-butadiene	mg/L	0.05	0.016	32	10-92	
Hexachlorobenzene	mg/L	0.05	0.030	60	34-89	
Hexachloroethane	mg/L	0.05	0.018	36	14-84	
Nitrobenzene	mg/L	0.05	0.031	63	39-113	
Pentachlorophenol	mg/L	0.05	0.033J	65	16-133	
Pyridine	mg/L	0.05	0.010	21	10-61	
2,4,6-Tribromophenol (S)	%			90	37-127	
2-Fluorobiphenyl (S)	%			51	36-92	
2-Fluorophenol (S)	%			45	23-69	
Nitrobenzene-d5 (S)	%			60	44-107	
p-Terphenyl-d14 (S)	%			78	57-146	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

LABORATORY CONTROL SAMPLE: 3466855

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenol-d5 (S)	%.			29	15-47	

MATRIX SPIKE SAMPLE: 3466856

Parameter	Units	50355005001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.050	0.5	0.23	46	18-91	
2,4,5-Trichlorophenol	mg/L	<0.050	0.5	0.40J	80	51-131	
2,4,6-Trichlorophenol	mg/L	<0.050	0.5	0.40	80	50-124	
2,4-Dinitrotoluene	mg/L	<0.050	0.5	0.37	74	41-118	
2-Methylphenol(o-Cresol)	mg/L	<0.050	0.5	0.24	48	23-115	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.10	1	0.57	57	18-110	
Hexachloro-1,3-butadiene	mg/L	<0.050	0.5	0.21	42	10-95	
Hexachlorobenzene	mg/L	<0.050	0.5	0.32	63	13-107	
Hexachloroethane	mg/L	<0.050	0.5	0.21	42	10-88	
Nitrobenzene	mg/L	<0.050	0.5	<0.050	0	36-113	M1
Pentachlorophenol	mg/L	<0.25	0.5	0.48J	97	27-139	
Pyridine	mg/L	<0.10	0.5	0.21	42	11-66	
2,4,6-Tribromophenol (S)	%.				118	37-127	
2-Fluorobiphenyl (S)	%.				63	36-92	
2-Fluorophenol (S)	%.				50	23-69	
Nitrobenzene-d5 (S)	%.				0	44-107	S0
p-Terphenyl-d14 (S)	%.				86	57-146	
Phenol-d5 (S)	%.				21	15-47	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50355005

QC Batch:	755322	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	Dry Weight/Percent Moisture
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

SAMPLE DUPLICATE: 3461821

Parameter	Units	50355005001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	99.0	98.9	0	10	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50355005

QC Batch:	755323	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	2540G Total Solids
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

SAMPLE DUPLICATE: 3461823

Parameter	Units	50355005001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	1.0	1.1	6	10	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50355005

QC Batch:	755324	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	2540G Total Volatile Solids
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

SAMPLE DUPLICATE: 3461824

Parameter	Units	50355005001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	54.1	52.9	2	10	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 755914

Analysis Method: EPA 9038

QC Batch Method: EPA 9038

Analysis Description: 9038 Sulfate Soil

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3464410

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Sulfate	mg/kg	<50.0	100	50.0	10/05/23 15:11	N2

LABORATORY CONTROL SAMPLE: 3464411

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Sulfate	mg/kg	200	192	96	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3464412 3464413

Parameter	Units	50355009001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Sulfate	mg/kg	157	608	608	792	782	105	103	90-110	1	20	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 756472

Analysis Method: EPA 9045

QC Batch Method: EPA 9045

Analysis Description: 9045 pH

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

SAMPLE DUPLICATE: 3466890

Parameter	Units	50355389001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	6.6	6.6	0	2	H3

SAMPLE DUPLICATE: 3466891

Parameter	Units	50355390001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	7.4	7.4	0	2	H3

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 756665

Analysis Method: EPA 351.2

QC Batch Method: EPA 351.2

Analysis Description: 351.2 TKN

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3467717

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<34.4	48.5	34.4	10/11/23 12:32	N2

LABORATORY CONTROL SAMPLE: 3467718

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	495	485	98	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3467719 3467720

Parameter	Units	60437996001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	37800	9860	9800	46200	48300	84	107	90-110	5	20	M0,N2

MATRIX SPIKE SAMPLE: 3467721

Parameter	Units	50355319001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	30500	2540	29300	-48	90-110	M0,N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50355005

QC Batch:	756630	Analysis Method:	EPA 353.2
QC Batch Method:	EPA 353.2	Analysis Description:	353.2 Nitrate + Nitrite
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3467510 Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<2.5	5.0	2.5	10/10/23 20:46	N2
Nitrogen, NO2 plus NO3	mg/kg	<2.5	5.0	2.5	10/10/23 20:46	N2

LABORATORY CONTROL SAMPLE: 3467511

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	10	10.3	103	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	20	20.2	101	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3467512 3467513

Parameter	Units	50355585001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	ND	112	112	111	113	98	100	90-110	2	20	N2
Nitrogen, NO2 plus NO3	mg/kg	ND	223	223	222	224	98	99	90-110	1	20	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 756615

Analysis Method: SM 4500-NH3 G

QC Batch Method: SM 4500-NH3 B

Analysis Description: 4500 Ammonia, Distilled

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3467434

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<2.3	4.9	2.3	10/11/23 17:42	N2

LABORATORY CONTROL SAMPLE: 3467435

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	19.8	20.0	101	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3467436 3467437

Parameter	Units	50355005001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	98400	17700	17000	115000	119000	95	118	90-110	3	20	M0, N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 757184

Analysis Method: SM 4500-Cl-E

QC Batch Method: SM 4500-Cl-E

Analysis Description: 4500 Chloride

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3470156

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Chloride	mg/kg	<49.8	99.6	49.8	10/13/23 13:50	N2

LABORATORY CONTROL SAMPLE: 3470157

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	199	209	105	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3470158 3470159

Parameter	Units	50355005001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	15200	19700	19700	41000	41300	131	133	90-110	1	20	M3, N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50355005

QC Batch: 756896

Analysis Method: SM 4500-CN-E

QC Batch Method: SM 4500-CN-C

Analysis Description: 4500CNE Cyanide

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50355005001

METHOD BLANK: 3468983

Matrix: Solid

Associated Lab Samples: 50355005001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Cyanide	mg/kg	<0.26	0.49	0.26	10/12/23 12:09	N2

LABORATORY CONTROL SAMPLE: 3468984

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/kg	4.9	5.0	103	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3468985 3468986

Parameter	Units	50355005001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Cyanide	mg/kg	<25.1	475	475	478	459	100	97	90-110	4	20	N2

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QUALIFIERS

Project: Biosolids
Pace Project No.: 50355005

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	A matrix spike/matrix spike duplicate was not performed for this batch due to insufficient sample extract volume.
2d	Instrument response was outside of Pace Analytical acceptance limits. The result is estimated. MTM 10/12/23
F5	The recovery of the analyte in the CRDL standard (also known as the reporting limit verification) did not meet the acceptance criteria.
H3	Sample was received or analysis requested beyond the recognized method holding time.
L1	Analyte recovery in the laboratory control sample (LCS) was above QC limits. Results for this analyte in associated samples may be biased high.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M1	Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
M3	Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
PO	The reported result is outside the range of the pH buffer solutions used to check the calibration of the pH meter.
R1	RPD value was outside control limits.
S0	Surrogate recovery outside laboratory control limits.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids

Pace Project No.: 50355005

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50355005001	Digester #4	EPA 3546	756328	EPA 8081	756394
50355005001	Digester #4	EPA 3510	756371	EPA 8081	756549
50355005001	Digester #4	EPA 3546	755208	EPA 8082	755286
50355005001	Digester #4	EPA 8151	756345	EPA 8151A	756644
50355005001	Digester #4	EPA 3050	755107	EPA 6010	756960
50355005001	Digester #4	EPA 3010	755887	EPA 6010	756110
50355005001	Digester #4	EPA 3050B	755713	EPA 6020	756159
50355005001	Digester #4	EPA 7470	755883	EPA 7470	756007
50355005001	Digester #4	EPA 7471	756431	EPA 7471	756589
50355005001	Digester #4	EPA 3546	755461	EPA 8270	756148
50355005001	Digester #4	EPA 3510	756452	EPA 8270	756627
50355005001	Digester #4	EPA 8260	755799		
50355005001	Digester #4	EPA 5030/8260	757059		
50355005001	Digester #4	SM 2540G	755322		
50355005001	Digester #4	SM 2540G	755323		
50355005001	Digester #4	SM 2540G	755324		
50355005001	Digester #4	EPA 9038	755914	EPA 9038	755958
50355005001	Digester #4	EPA 9045	756472		
50355005001	Digester #4	EPA 351.2	756665	EPA 351.2	756710
50355005001	Digester #4	EPA 353.2	756630	EPA 353.2	756631
50355005001	Digester #4	SM 4500-NH3 B	756615	SM 4500-NH3 G	756835
50355005001	Digester #4	SM 4500-CI-E	757184	SM 4500-CI-E	757195
50355005001	Digester #4	SM 4500-CN-C	756896	SM 4500-CN-E	757035

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CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

Company Name: City of Cadillac Street Address: 1121 Pieltz Rd, Cadillac, MI 49601	Contact/Report To: Tomaszewski, Cindy Phone #: 231-775-2368 E-Mail: ctomaszewski@cadillac-mi.net Cc E-Mail:
Customer Project #: Project Name: Biosolids	Invoice To: Invoice E-Mail:
Site Collection Info/Facility ID (as applicable): WWTP	Purchase Order # (if applicable): Quote #:
Time Zone Collected: [] AK [] PT [] MT [] CT [] ET	County / State origin of sample(s): Michigan
Data Deliverables: [] Level II [] Level III [] Level IV [] EQUIS [] Other _____	Regulatory Program (DW, RCRA, etc.) as applicable: NPDES Rush (Pre-approval required): [] 2 Day [] 3 day [] 5 day [] Other _____ Date Results Requested: _____
	DW PWSID # or WW Permit # as applicable: Field Filtered (if applicable): [] Yes ☒ No Analysis:

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Waste Water (WW), Product (P), Soil/Solid (SS), Oil (OL), Wipe (WP), Tissue (TS), Biosassay (B), Vapor (V), Other (OT), Surface Water (SW), Sediment (SED), Sludge (SL), Caulk

[illegible]

Customer Remarks / Special Conditions / Possible Hazards:	Collected By:
Wet Chem: TKN, NH ₃ , CN, Cl, NO ₂ /NO ₃ , SO ₄ , pH, %Solid, TVS	Printed Name: CHAP/KEVIN Signature: 
Relinquished by/Company: (Signature)	Received by/Company: (Signature)
	
Relinquished by/Company: (Signature)	Received by/Company: (Signature)
Relinquished by/Company: (Signature)	Received by/Company: (Signature)
Relinquished by/Company: (Signature)	Received by/Company: (Signature)

Page 52



Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>9.29.23</u>		Evaluated By: <u>WDC</u>		WO#: 50355005 PM: BJH Due Date: 10/13/23 CLIENT: GR-Cadillac	
Client: <u>City of Cadillac</u>		PM: <u>BJH</u>			
Lab Notified of Rush or Short Holds: YES NO NA					
Project Received Via: FedEx ☒ UPS ☒ Client ☐ Pace Courier ☐ Other: _____				Comments:	
Custody Seal Present and Intact:		YES	NO	NA	
Received Sample Information Form (SIF): Drinking Waters Only		YES	NO	NA	
Short Hold Present (≤ 48 Hours):		YES	NO		
Sample Received in Hold:		YES	NO		
Custody Signature Present:		YES	NO		
Collector Signature Present:		YES	NO		
Sample Collected Today and On Ice:		YES	NO	N/A	
IR Gun #: <u>350</u> <u>351</u> Therm #: <u>282</u> <u>283</u>		Temp. should be 0°C - 6°C (Initial/Corrected)			
Ice Type: <u>WET Bagged / WET Loose</u> <u>BLUE</u> <u>NONE</u>		1. Cooler Temp. Upon Receipt: <u>3.9/3.0</u> °C			
Ice Location: <u>TOP</u> <u>BOTTOM</u> <u>MIDDLE</u> <u>DISPERSED</u>		2. Cooler Temp. Upon Receipt: _____ °C			
Temp Blank Received:		YES	NO		
Sample Label Matches COC (ID/Date/Time):		YES	NO		
Container Intact:		YES	NO		
Correct Container:		YES	NO		
Sufficient Volume:		YES	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. pH Strip Lot #: _____ Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl		YES	NO	N/A	
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide		YES	NO	N/A	
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.		YES	NO	N/A	
Trip Blank Received: HCl MeOH Other: _____		YES	NO	ON HOLD	
Comments:		3. Cooler Temp. Upon Receipt: _____ °C			
		4. Cooler Temp. Upon Receipt: _____ °C			
		Non-Conformance Form Required: YES NO			